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Book Author(s): James D. Hamilton

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Covariance-Stationary Vector Processes

This is the first of two chapters introducing vector time series. Chapter 10 is devoted to the theory of multivariate dynamic systems, while Chapter 11 focuses on empirical issues of estimating and interpreting vector autoregressions. Only the first section of Chapter 10 is necessary for understanding the material in Chapter 11.

Section 10.1 introduces some of the key ideas in vector time series analysis. Section 10.2 develops some convergence results that are useful for deriving the asymptotic properties of certain statistics and for characterizing the consequences of multivariate filters. Section 10.3 introduces the autocovariance-generating function for vector processes, which is used to analyze the multivariate spectrum in Section 10.4. Section 10.5 develops a multivariate generalization of Proposition 7.5, describing the asymptotic properties of the sample mean of a serially correlated vector process. These last results are useful for deriving autocorrelation- and heteroskedasticity-consistent estimators for *OLS*, for understanding the properties of generalized method of moments estimators discussed in Chapter 14, and for deriving some of the tests for unit roots discussed in Chapter 17.

10.1. Introduction to Vector Autoregressions

Chapter 3 proposed modeling a scalar time series y_t in terms of an autoregression:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad [10.1.1]$$

where

$$E(\varepsilon_t) = 0 \quad [10.1.2]$$

$$E(\varepsilon_t \varepsilon_\tau) = \begin{cases} \sigma^2 & \text{for } t = \tau \\ 0 & \text{otherwise.} \end{cases} \quad [10.1.3]$$

Note that we will continue to use the convention introduced in Chapter 8 of using lowercase letters to denote either a random variable or its realization. This chapter describes the dynamic interactions among a set of variables collected in an $(n \times 1)$ vector $\mathbf{y}_t$. For example, the first element of $\mathbf{y}_t$ (denoted y_{1t}) might represent the level of GNP in year t , the second element (y_{2t}) the interest rate paid on Treasury bills in year t , and so on. A *p*th-order vector autoregression, denoted $VAR(p)$, is a vector generalization of [10.1.1] through [10.1.3]:

$$\mathbf{y}_t = \mathbf{c} + \Phi_1 \mathbf{y}_{t-1} + \Phi_2 \mathbf{y}_{t-2} + \cdots + \Phi_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t. \quad [10.1.4]$$

Here $\mathbf{c}$ denotes an $(n \times 1)$ vector of constants and Φ_j an $(n \times n)$ matrix of autoregressive coefficients for $j = 1, 2, \dots, p$. The $(n \times 1)$ vector $\boldsymbol{\varepsilon}_t$ is a vector

generalization of white noise:

$$E(\boldsymbol{\varepsilon}_t) = \mathbf{0} \quad [10.1.5]$$

$$E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_\tau') = \begin{cases} \boldsymbol{\Omega} & \text{for } t = \tau \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad [10.1.6]$$

with $\boldsymbol{\Omega}$ an $(n \times n)$ symmetric positive definite matrix.

Let c_i denote the i th element of the vector $\mathbf{c}$ and let $\phi_{ij}^{(1)}$ denote the row i , column j element of the matrix Φ_1 . Then the first row of the vector system in [10.1.4] specifies that

$$\begin{aligned} y_{1t} = & c_1 + \phi_{11}^{(1)} y_{1,t-1} + \phi_{12}^{(1)} y_{2,t-1} + \cdots + \phi_{1n}^{(1)} y_{n,t-1} \\ & + \phi_{11}^{(2)} y_{1,t-2} + \phi_{12}^{(2)} y_{2,t-2} + \cdots + \phi_{1n}^{(2)} y_{n,t-2} \\ & + \cdots + \phi_{11}^{(p)} y_{1,t-p} + \phi_{12}^{(p)} y_{2,t-p} + \cdots + \phi_{1n}^{(p)} y_{n,t-p} + \varepsilon_{1t}. \end{aligned} \quad [10.1.7]$$

Thus, a vector autoregression is a system in which each variable is regressed on a constant and p of its own lags as well as on p lags of each of the other variables in the VAR. Note that each regression has the same explanatory variables.

Using lag operator notation, [10.1.4] can be written in the form

$$[\mathbf{I}_n - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p] \mathbf{y}_t = \mathbf{c} + \boldsymbol{\varepsilon}_t$$

or

$$\Phi(L) \mathbf{y}_t = \mathbf{c} + \boldsymbol{\varepsilon}_t.$$

Here $\Phi(L)$ indicates an $(n \times n)$ matrix polynomial in the lag operator L . The row i , column j element of $\Phi(L)$ is a scalar polynomial in L :

$$\Phi(L) = [\delta_{ij} - \phi_{ij}^{(1)} L^1 - \phi_{ij}^{(2)} L^2 - \cdots - \phi_{ij}^{(p)} L^p],$$

where δ_{ij} is unity if $i = j$ and zero otherwise.

A vector process $\mathbf{y}_t$ is said to be covariance-stationary if its first and second moments ($E[\mathbf{y}_t]$ and $E[\mathbf{y}_t \mathbf{y}_{t-j}']$, respectively) are independent of the date t . If the process is covariance-stationary, we can take expectations of both sides of [10.1.4] to calculate the mean $\boldsymbol{\mu}$ of the process:

$$\boldsymbol{\mu} = \mathbf{c} + \Phi_1 \boldsymbol{\mu} + \Phi_2 \boldsymbol{\mu} + \cdots + \Phi_p \boldsymbol{\mu},$$

or

$$\boldsymbol{\mu} = (\mathbf{I}_n - \Phi_1 - \Phi_2 - \cdots - \Phi_p)^{-1} \mathbf{c}.$$

Equation [10.1.4] can then be written in terms of deviations from the mean as

$$\begin{aligned} (\mathbf{y}_t - \boldsymbol{\mu}) = & \Phi_1 (\mathbf{y}_{t-1} - \boldsymbol{\mu}) \\ & + \Phi_2 (\mathbf{y}_{t-2} - \boldsymbol{\mu}) + \cdots + \Phi_p (\mathbf{y}_{t-p} - \boldsymbol{\mu}) + \boldsymbol{\varepsilon}_t. \end{aligned} \quad [10.1.8]$$

Rewriting a VAR(p) as a VAR(1)

As in the case of the univariate $AR(p)$ process, it is helpful to rewrite [10.1.8] in terms of a $VAR(1)$ process. Toward this end, define

$$\underset{(np \times 1)}{\xi_t} \equiv \begin{bmatrix} \mathbf{y}_t - \boldsymbol{\mu} \\ \mathbf{y}_{t-1} - \boldsymbol{\mu} \\ \vdots \\ \mathbf{y}_{t-p+1} - \boldsymbol{\mu} \end{bmatrix} \quad [10.1.9]$$

$$\underset{(np \times np)}{\mathbf{F}} \equiv \begin{bmatrix} \Phi_1 & \Phi_2 & \Phi_3 & \cdots & \Phi_{p-1} & \Phi_p \\ \mathbf{I}_n & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_n & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_n & \mathbf{0} \end{bmatrix} \quad [10.1.10]$$

$$\underset{(np \times 1)}{\mathbf{v}_t} \equiv \begin{bmatrix} \boldsymbol{\varepsilon}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}.$$

The $VAR(p)$ in [10.1.8] can then be rewritten as the following $VAR(1)$:

$$\xi_t = \mathbf{F}\xi_{t-1} + \mathbf{v}_t, \quad [10.1.11]$$

where

$$E(\mathbf{v}_t \mathbf{v}_t') = \begin{cases} \mathbf{Q} & \text{for } t = \tau \\ \mathbf{0} & \text{otherwise} \end{cases}$$

and

$$\underset{(np \times np)}{\mathbf{Q}} \equiv \begin{bmatrix} \boldsymbol{\Omega} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix}$$

Conditions for Stationarity

Equation [10.1.11] implies that

$$\xi_{t+s} = \mathbf{v}_{t+s} + \mathbf{F}\mathbf{v}_{t+s-1} + \mathbf{F}^2\mathbf{v}_{t+s-2} + \cdots + \mathbf{F}^{s-1}\mathbf{v}_{t+1} + \mathbf{F}^s\xi_t. \quad [10.1.12]$$

In order for the process to be covariance-stationary, the consequences of any given $\boldsymbol{\varepsilon}_t$ must eventually die out. If the eigenvalues of $\mathbf{F}$ all lie inside the unit circle, then the VAR turns out to be covariance-stationary.

The following result generalizes Proposition 1.1 from Chapter 1 (for a proof see Appendix 10.A at the end of this chapter).

Proposition 10.1: *The eigenvalues of the matrix $\mathbf{F}$ in [10.1.10] satisfy*

$$|\mathbf{I}_n \lambda^p - \Phi_1 \lambda^{p-1} - \Phi_2 \lambda^{p-2} - \cdots - \Phi_p| = 0. \quad [10.1.13]$$

Hence, a $VAR(p)$ is covariance-stationary as long as $|\lambda| < 1$ for all values of λ satisfying [10.1.13]. Equivalently, the VAR is covariance-stationary if all values of z satisfying

$$|\mathbf{I}_n - \Phi_1 z - \Phi_2 z^2 - \cdots - \Phi_p z^p| = 0$$

lie outside the unit circle.

Vector $MA(\infty)$ Representation

The first n rows of the vector system represented in [10.1.12] constitute a vector generalization of equation [4.2.20]:

$$\mathbf{y}_{t+s} = \boldsymbol{\mu} + \boldsymbol{\epsilon}_{t+s} + \boldsymbol{\Psi}_1 \boldsymbol{\epsilon}_{t+s-1} + \boldsymbol{\Psi}_2 \boldsymbol{\epsilon}_{t+s-2} + \cdots + \boldsymbol{\Psi}_{s-1} \boldsymbol{\epsilon}_{t+1} \\ + \mathbf{F}_{11}^{(s)}(\mathbf{y}_t - \boldsymbol{\mu}) + \mathbf{F}_{12}^{(s)}(\mathbf{y}_{t-1} - \boldsymbol{\mu}) + \cdots + \mathbf{F}_{1p}^{(s)}(\mathbf{y}_{t-p+1} - \boldsymbol{\mu}). \quad [10.1.14]$$

Here $\boldsymbol{\Psi}_j = \mathbf{F}_{11}^{(j)}$ and $\mathbf{F}_{11}^{(j)}$ denotes the upper left block of $\mathbf{F}^j$, where $\mathbf{F}^j$ is the matrix $\mathbf{F}$ raised to the j th power—that is, the $(n \times n)$ matrix $\mathbf{F}_{11}^{(j)}$ indicates rows 1 through n and columns 1 through n of the $(np \times np)$ matrix $\mathbf{F}^j$. Similarly, $\mathbf{F}_{12}^{(j)}$ denotes the block of $\mathbf{F}^j$ consisting of rows 1 through n and columns $(n+1)$ through $2n$, while $\mathbf{F}_{1p}^{(j)}$ denotes rows 1 through n and columns $[n(p-1)+1]$ through np of $\mathbf{F}^j$.

If the eigenvalues of $\mathbf{F}$ all lie inside the unit circle, then $\mathbf{F}^s \rightarrow \mathbf{0}$ as $s \rightarrow \infty$ and $\mathbf{y}_t$ can be expressed as a convergent sum of the history of $\boldsymbol{\epsilon}$:

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\epsilon}_t + \boldsymbol{\Psi}_1 \boldsymbol{\epsilon}_{t-1} + \boldsymbol{\Psi}_2 \boldsymbol{\epsilon}_{t-2} + \boldsymbol{\Psi}_3 \boldsymbol{\epsilon}_{t-3} + \cdots \equiv \boldsymbol{\mu} + \boldsymbol{\Psi}(L) \boldsymbol{\epsilon}_t, \quad [10.1.15]$$

which is a vector $MA(\infty)$ representation.

Note that $\mathbf{y}_{t-j}$ is a linear function of $\boldsymbol{\epsilon}_{t-j}, \boldsymbol{\epsilon}_{t-j-1}, \dots$, each of which is uncorrelated with $\boldsymbol{\epsilon}_{t+1}$ for $j = 0, 1, \dots$. It follows that $\boldsymbol{\epsilon}_{t+1}$ is uncorrelated with $\mathbf{y}_{t-j}$ for any $j \geq 0$. Thus, the linear forecast of $\mathbf{y}_{t+1}$ on the basis of $\mathbf{y}_t, \mathbf{y}_{t-1}, \dots$ is given by

$$\hat{\mathbf{y}}_{t+1|t} = \boldsymbol{\mu} + \boldsymbol{\Phi}_1(\mathbf{y}_t - \boldsymbol{\mu}) + \boldsymbol{\Phi}_2(\mathbf{y}_{t-1} - \boldsymbol{\mu}) + \cdots + \boldsymbol{\Phi}_p(\mathbf{y}_{t-p+1} - \boldsymbol{\mu}),$$

and $\boldsymbol{\epsilon}_{t+1}$ can be interpreted as the fundamental innovation for $\mathbf{y}_{t+1}$, that is, the error in forecasting $\mathbf{y}_{t+1}$ on the basis of a linear function of a constant and $\mathbf{y}_t, \mathbf{y}_{t-1}, \dots$. More generally, it follows from [10.1.14] that a forecast of $\mathbf{y}_{t+s}$ on the basis of $\mathbf{y}_t, \mathbf{y}_{t-1}, \dots$ will take the form

$$\hat{\mathbf{y}}_{t+s|t} = \boldsymbol{\mu} + \mathbf{F}_{11}^{(s)}(\mathbf{y}_t - \boldsymbol{\mu}) + \mathbf{F}_{12}^{(s)}(\mathbf{y}_{t-1} - \boldsymbol{\mu}) \\ + \cdots + \mathbf{F}_{1p}^{(s)}(\mathbf{y}_{t-p+1} - \boldsymbol{\mu}). \quad [10.1.16]$$

The moving average matrices $\boldsymbol{\Psi}_j$ could equivalently be calculated as follows. The operators $\boldsymbol{\Phi}(L)$ and $\boldsymbol{\Psi}(L)$ are related by

$$\boldsymbol{\Psi}(L) = [\boldsymbol{\Phi}(L)]^{-1},$$

requiring

$$[\mathbf{I}_n - \boldsymbol{\Phi}_1 L - \boldsymbol{\Phi}_2 L^2 - \cdots - \boldsymbol{\Phi}_p L^p][\mathbf{I}_n + \boldsymbol{\Psi}_1 L + \boldsymbol{\Psi}_2 L^2 + \cdots] = \mathbf{I}_n.$$

Setting the coefficient on L^1 equal to the zero matrix, as in Exercise 3.3 of Chapter 3, produces

$$\boldsymbol{\Psi}_1 - \boldsymbol{\Phi}_1 = \mathbf{0}. \quad [10.1.17]$$

Similarly, setting the coefficient on L^2 equal to zero gives

$$\boldsymbol{\Psi}_2 = \boldsymbol{\Phi}_1 \boldsymbol{\Psi}_1 + \boldsymbol{\Phi}_2, \quad [10.1.18]$$

and in general for L^s ,

$$\boldsymbol{\Psi}_s = \boldsymbol{\Phi}_1 \boldsymbol{\Psi}_{s-1} + \boldsymbol{\Phi}_2 \boldsymbol{\Psi}_{s-2} + \cdots + \boldsymbol{\Phi}_p \boldsymbol{\Psi}_{s-p} \quad \text{for } s = 1, 2, \dots, \quad [10.1.19]$$

with $\boldsymbol{\Psi}_0 = \mathbf{I}_n$ and $\boldsymbol{\Psi}_s = \mathbf{0}$ for $s < 0$.

Note that the innovation in the $MA(\infty)$ representation [10.1.15] is $\boldsymbol{\epsilon}_t$, the fundamental innovation for $\mathbf{y}$. There are alternative moving average representations based on vector white noise processes other than $\boldsymbol{\epsilon}_t$. Let $\mathbf{H}$ denote a nonsingular

($n \times n$) matrix, and define

$$\mathbf{u}_t \equiv \mathbf{H}\boldsymbol{\varepsilon}_t. \quad [10.1.20]$$

Then certainly $\mathbf{u}_t$ is white noise. Moreover, from [10.1.15] we could write

$$\begin{aligned} \mathbf{y}_t &= \boldsymbol{\mu} + \mathbf{H}^{-1}\mathbf{H}\boldsymbol{\varepsilon}_t + \boldsymbol{\Psi}_1\mathbf{H}^{-1}\mathbf{H}\boldsymbol{\varepsilon}_{t-1} + \boldsymbol{\Psi}_2\mathbf{H}^{-1}\mathbf{H}\boldsymbol{\varepsilon}_{t-2} \\ &\quad + \boldsymbol{\Psi}_3\mathbf{H}^{-1}\mathbf{H}\boldsymbol{\varepsilon}_{t-3} + \cdots \\ &= \boldsymbol{\mu} + \mathbf{J}_0\mathbf{u}_t + \mathbf{J}_1\mathbf{u}_{t-1} + \mathbf{J}_2\mathbf{u}_{t-2} + \mathbf{J}_3\mathbf{u}_{t-3} + \cdots, \end{aligned} \quad [10.1.21]$$

where

$$\mathbf{J}_s \equiv \boldsymbol{\Psi}_s\mathbf{H}^{-1}.$$

For example, $\mathbf{H}$ could be any matrix that diagonalizes $\boldsymbol{\Omega}$, the variance-covariance matrix of $\boldsymbol{\varepsilon}_t$:

$$\mathbf{H}\boldsymbol{\Omega}\mathbf{H}' = \mathbf{D},$$

with $\mathbf{D}$ a diagonal matrix. For such a choice of $\mathbf{H}$, the elements of $\mathbf{u}_t$ are uncorrelated with one another:

$$E(\mathbf{u}_t\mathbf{u}_t') = E(\mathbf{H}\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t'\mathbf{H}') = \mathbf{D}.$$

Thus, it is always possible to write a stationary $VAR(p)$ process as a convergent infinite moving average of a white noise vector $\mathbf{u}_t$ whose elements are mutually uncorrelated.

There is one important difference between the $MA(\infty)$ representations [10.1.15] and [10.1.21], however. In [10.1.15], the leading MA parameter matrix ($\boldsymbol{\Psi}_0$) is the identity matrix, whereas in [10.1.21] the leading MA parameter matrix ($\mathbf{J}_0$) is not the identity matrix. To obtain the MA representation for the fundamental innovations, we must impose the normalization $\boldsymbol{\Psi}_0 = \mathbf{I}_n$.

Assumptions Implicit in a VAR

For a covariance-stationary process, the parameters $\mathbf{c}$ and $\boldsymbol{\Phi}_1, \dots, \boldsymbol{\Phi}_p$ in equation [10.1.4] could be defined as the coefficients of the projection of $\mathbf{y}_t$ on a constant and $\mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-p}$. Thus, $\boldsymbol{\varepsilon}_t$ is uncorrelated with $\mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-p}$ by the definition of $\boldsymbol{\Phi}_1, \dots, \boldsymbol{\Phi}_p$. The parameters of a vector autoregression can accordingly be estimated consistently with n *OLS* regressions of the form of [10.1.7]. The additional assumption implicit in a VAR is that the $\boldsymbol{\varepsilon}_t$ defined by this projection is further uncorrelated with $\mathbf{y}_{t-p-1}, \mathbf{y}_{t-p-2}, \dots$. The assumption that $\mathbf{y}_t$ follows a vector autoregression is basically the assumption that p lags are sufficient to summarize all of the dynamic correlations between elements of $\mathbf{y}$.

10.2. Autocovariances and Convergence Results for Vector Processes

The j th Autocovariance Matrix

For a covariance-stationary n -dimensional vector process, the j th autocovariance is defined to be the following ($n \times n$) matrix:

$$\boldsymbol{\Gamma}_j = E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-j} - \boldsymbol{\mu})']. \quad [10.2.1]$$

Note that although $\gamma_j = \gamma_{-j}$ for a scalar process, the same is not true of a vector

process:

$$\Gamma_j \neq \Gamma_{-j}.$$

For example, the (1, 2) element of Γ_j gives the covariance between y_{1t} and $y_{2,t-j}$. The (1, 2) element of Γ_{-j} gives the covariance between y_{1t} and $y_{2,t+j}$. There is no reason that these should be related—the response of y_1 to previous movements in y_2 could be completely different from the response of y_2 to previous movements in y_1 .

Instead, the correct relation is

$$\Gamma'_j = \Gamma_{-j}. \quad [10.2.2]$$

To derive [10.2.2], notice that covariance-stationarity would mean that t in [10.2.1] could be replaced with $t + j$:

$$\Gamma_j = E[(y_{t+j} - \mu)(y_{t+j-j} - \mu)'] = E[(y_{t+j} - \mu)(y_t - \mu)'].$$

Taking transposes,

$$\Gamma'_j = E[(y_t - \mu)(y_{t+j} - \mu)'] = \Gamma_{-j},$$

as claimed.

Vector MA(q) Process

A vector moving average process of order q takes the form

$$y_t = \mu + \epsilon_t + \Theta_1 \epsilon_{t-1} + \Theta_2 \epsilon_{t-2} + \cdots + \Theta_q \epsilon_{t-q}, \quad [10.2.3]$$

where ϵ_t is a vector white noise process satisfying [10.1.5] and [10.1.6] and Θ_j denotes an $(n \times n)$ matrix of MA coefficients for $j = 1, 2, \dots, q$. The mean of y_t is μ , and the variance is

$$\begin{aligned} \Gamma_0 &= E[(y_t - \mu)(y_t - \mu)'] \\ &= E[\epsilon_t \epsilon_t'] + \Theta_1 E[\epsilon_{t-1} \epsilon_{t-1}'] \Theta_1' + \Theta_2 E[\epsilon_{t-2} \epsilon_{t-2}'] \Theta_2' \\ &\quad + \cdots + \Theta_q E[\epsilon_{t-q} \epsilon_{t-q}'] \Theta_q' \\ &= \Omega + \Theta_1 \Omega \Theta_1' + \Theta_2 \Omega \Theta_2' + \cdots + \Theta_q \Omega \Theta_q', \end{aligned} \quad [10.2.4]$$

with autocovariances

$$\Gamma_j = \begin{cases} \Theta_j \Omega + \Theta_{j+1} \Omega \Theta_1' + \Theta_{j+2} \Omega \Theta_2' + \cdots + \Theta_q \Omega \Theta_{q-j}' & \text{for } j = 1, 2, \dots, q \\ \Omega \Theta_{-j}' + \Theta_1 \Omega \Theta_{-j+1}' + \Theta_2 \Omega \Theta_{-j+2}' + \cdots + \Theta_{q+j} \Omega \Theta_q' & \text{for } j = -1, -2, \dots, -q \\ 0 & \text{for } |j| > q, \end{cases} \quad [10.2.5]$$

where $\Theta_0 = \mathbf{I}_n$. Thus, any vector MA(q) process is covariance-stationary.

Vector MA(∞) Process

The vector MA(∞) process is written

$$y_t = \mu + \epsilon_t + \Psi_1 \epsilon_{t-1} + \Psi_2 \epsilon_{t-2} + \cdots \quad [10.2.6]$$

for ϵ_t again satisfying [10.1.5] and [10.1.6].

A sequence of scalars $\{h_s\}_{s=-\infty}^{\infty}$ was said to be absolutely summable if $\sum_{s=-\infty}^{\infty} |h_s| < \infty$. For $\mathbf{H}_s$ an $(n \times m)$ matrix, the sequence of matrices $\{\mathbf{H}_s\}_{s=-\infty}^{\infty}$ is

absolutely summable if each of its elements forms an absolutely summable scalar sequence. For example, if $\psi_{ij}^{(s)}$ denotes the row i , column j element of the moving average parameter matrix Ψ_s associated with lag s , then the sequence $\{\Psi_s\}_{s=0}^\infty$ is absolutely summable if

$$\sum_{s=0}^{\infty} |\psi_{ij}^{(s)}| < \infty \quad \text{for } i = 1, 2, \dots, n \text{ and } j = 1, 2, \dots, n. \quad [10.2.7]$$

Many of the results for scalar $MA(\infty)$ processes with absolutely summable coefficients go through for vector processes as well. This is summarized by the following theorem, proved in Appendix 10.A to this chapter.

Proposition 10.2: Let $\mathbf{y}_t$ be an $(n \times 1)$ vector satisfying

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{k=0}^{\infty} \Psi_k \boldsymbol{\epsilon}_{t-k},$$

where $\boldsymbol{\epsilon}_t$ is vector white noise satisfying [10.1.5] and [10.1.6] and $\{\Psi_k\}_{k=0}^\infty$ is absolutely summable. Let y_{it} denote the i th element of $\mathbf{y}_t$, and let μ_i denote the i th element of $\boldsymbol{\mu}$. Then

- (a) the autocovariance between the i th variable at time t and the j th variable s periods earlier, $E(y_{it} - \mu_i)(y_{j,t-s} - \mu_j)$, exists and is given by the row i , column j element of

$$\Gamma_s = \sum_{v=0}^{\infty} \Psi_{s+v} \Omega \Psi_v' \quad \text{for } s = 0, 1, 2, \dots;$$

- (b) the sequence of matrices $\{\Gamma_s\}_{s=0}^\infty$ is absolutely summable.

If, furthermore, $\{\boldsymbol{\epsilon}_t\}_{t=-\infty}^\infty$ is an i.i.d. sequence with $E|\epsilon_{i_1,t}\epsilon_{i_2,t}\epsilon_{i_3,t}\epsilon_{i_4,t}| < \infty$ for $i_1, i_2, i_3, i_4 = 1, 2, \dots, n$, then also,

- (c) $E|y_{i_1,t_1}y_{i_2,t_2}y_{i_3,t_3}y_{i_4,t_4}| < \infty$ for $i_1, i_2, i_3, i_4 = 1, 2, \dots, n$ and for all t_1, t_2, t_3, t_4 ;

- (d) $(1/T) \sum_{t=1}^T y_{it}y_{j,t-s} \xrightarrow{p} E(y_{it}y_{j,t-s})$ for $i, j = 1, 2, \dots, n$ and for all s .

Result (a) implies that the second moments of an $MA(\infty)$ vector process with absolutely summable coefficients can be found by taking the limit of [10.2.5] as $q \rightarrow \infty$. Result (b) is a convergence condition on these moments that will turn out to ensure that the vector process is ergodic for the mean (see Proposition 10.5 later in this chapter). Result (c) says that $\mathbf{y}_t$ has bounded fourth moments, while result (d) establishes that $\mathbf{y}_t$ is ergodic for second moments.

Note that the vector $MA(\infty)$ representation of a stationary vector autoregression calculated from [10.1.4] satisfies the absolute summability condition. To see this, recall from [10.1.14] that Ψ_s is a block of the matrix $\mathbf{F}^s$. If $\mathbf{F}$ has np distinct eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_{np})$, then any element of Ψ_s can be written as a weighted average of these eigenvalues as in equation [1.2.20];

$$\psi_{ij}^{(s)} = c_1(i, j) \cdot \lambda_1^s + c_2(i, j) \cdot \lambda_2^s + \dots + c_{np}(i, j) \cdot \lambda_{np}^s,$$

where $c_v(i, j)$ denotes a constant that depends on v, i , and j but not s . Absolute summability [10.2.7] then follows from the same arguments as in Exercise 3.5.

10.2. Autocovariances and Convergence Results for Vector Processes 263

Multivariate Filters

Suppose that the $(n \times 1)$ vector $\mathbf{y}_t$ follows an $MA(\infty)$ process:

$$\mathbf{y}_t = \boldsymbol{\mu}_Y + \boldsymbol{\Psi}(L)\boldsymbol{\epsilon}_t, \quad [10.2.8]$$

with $\{\boldsymbol{\Psi}_k\}_{k=0}^{\infty}$ absolutely summable. Let $\{\mathbf{H}_k\}_{k=-\infty}^{\infty}$ be an absolutely summable sequence of $(r \times n)$ matrices and suppose that an $(r \times 1)$ vector $\mathbf{x}_t$ is related to $\mathbf{y}_t$ according to

$$\mathbf{x}_t = \mathbf{H}(L)\mathbf{y}_t = \sum_{k=-\infty}^{\infty} \mathbf{H}_k \mathbf{y}_{t-k}. \quad [10.2.9]$$

That is,

$$\begin{aligned} \mathbf{x}_t &= \mathbf{H}(L)[\boldsymbol{\mu}_Y + \boldsymbol{\Psi}(L)\boldsymbol{\epsilon}_t] \\ &= \mathbf{H}(1)\boldsymbol{\mu}_Y + \mathbf{H}(L)\boldsymbol{\Psi}(L)\boldsymbol{\epsilon}_t \\ &= \boldsymbol{\mu}_X + \mathbf{B}(L)\boldsymbol{\epsilon}_t, \end{aligned} \quad [10.2.10]$$

where $\boldsymbol{\mu}_X \equiv \mathbf{H}(1)\boldsymbol{\mu}_Y$ and $\mathbf{B}(L)$ is the compound operator given by

$$\mathbf{B}(L) = \sum_{k=-\infty}^{\infty} \mathbf{B}_k L^k = \mathbf{H}(L)\boldsymbol{\Psi}(L). \quad [10.2.11]$$

The following proposition establishes that $\mathbf{x}_t$ follows an absolutely summable two-sided $MA(\infty)$ process.

Proposition 10.3: Let $\{\boldsymbol{\Psi}_k\}_{k=0}^{\infty}$ be an absolutely summable sequence of $(n \times n)$ matrices and let $\{\mathbf{H}_k\}_{k=-\infty}^{\infty}$ be an absolutely summable sequence of $(r \times n)$ matrices. Then the sequence of matrices $\{\mathbf{B}_k\}_{k=-\infty}^{\infty}$ associated with the operator $\mathbf{B}(L) = \mathbf{H}(L)\boldsymbol{\Psi}(L)$ is absolutely summable.

If $\{\boldsymbol{\epsilon}_t\}$ in [10.2.8] is i.i.d. with finite fourth moments, then $\{\mathbf{x}_t\}$ in [10.2.9] has finite fourth moments and is ergodic for second moments.

Vector Autoregression

Next we derive expressions for the second moments for $\mathbf{y}_t$ following a $VAR(p)$. Let $\boldsymbol{\xi}_t$ be as defined in equation [10.1.9]. Assuming that $\boldsymbol{\xi}$ and $\mathbf{y}$ are covariance-stationary, let $\boldsymbol{\Sigma}$ denote the variance of $\boldsymbol{\xi}_t$,

$$\begin{aligned} \boldsymbol{\Sigma} &= E(\boldsymbol{\xi}_t \boldsymbol{\xi}_t') \\ &= E \left\{ \begin{bmatrix} \mathbf{y}_t - \boldsymbol{\mu} \\ \mathbf{y}_{t-1} - \boldsymbol{\mu} \\ \vdots \\ \mathbf{y}_{t-p+1} - \boldsymbol{\mu} \end{bmatrix} \right. \\ &\quad \times \left. [(\mathbf{y}_t - \boldsymbol{\mu})' \quad (\mathbf{y}_{t-1} - \boldsymbol{\mu})' \quad \cdots \quad (\mathbf{y}_{t-p+1} - \boldsymbol{\mu})'] \right\} \\ &= \begin{bmatrix} \boldsymbol{\Gamma}_0 & \boldsymbol{\Gamma}_1 & \cdots & \boldsymbol{\Gamma}_{p-1} \\ \boldsymbol{\Gamma}_1' & \boldsymbol{\Gamma}_0 & \cdots & \boldsymbol{\Gamma}_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\Gamma}_{p-1}' & \boldsymbol{\Gamma}_{p-2}' & \cdots & \boldsymbol{\Gamma}_0 \end{bmatrix}, \end{aligned} \quad [10.2.12]$$

where Γ_j denotes the j th autocovariance of the original process $\mathbf{y}$. Postmultiplying [10.1.11] by its own transpose and taking expectations gives

$$E[\xi_t \xi_t'] = E[(\mathbf{F}\xi_{t-1} + \mathbf{v}_t)(\mathbf{F}\xi_{t-1} + \mathbf{v}_t)'] = \mathbf{F}E(\xi_{t-1}\xi_{t-1}')\mathbf{F}' + E(\mathbf{v}_t\mathbf{v}_t'),$$

or

$$\Sigma = \mathbf{F}\Sigma\mathbf{F}' + \mathbf{Q}. \quad [10.2.13]$$

A closed-form solution to [10.2.13] can be obtained in terms of the *vec* operator. If $\mathbf{A}$ is an $(m \times n)$ matrix, then $\text{vec}(\mathbf{A})$ is an $(mn \times 1)$ column vector, obtained by stacking the columns of $\mathbf{A}$, one below the other, with the columns ordered from left to right. For example, if

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix},$$

then

$$\text{vec}(\mathbf{A}) = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{12} \\ a_{22} \\ a_{32} \end{bmatrix}. \quad [10.2.14]$$

Appendix 10.A establishes the following useful result.

Proposition 10.4: *Let $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ be matrices whose dimensions are such that the product $\mathbf{ABC}$ exists. Then*

$$\text{vec}(\mathbf{ABC}) = (\mathbf{C}' \otimes \mathbf{A}) \cdot \text{vec}(\mathbf{B}) \quad [10.2.15]$$

where the symbol $\otimes$ denotes the Kronecker product.

Thus, if the *vec* operator is applied to both sides of [10.2.13], the result is

$$\text{vec}(\Sigma) = (\mathbf{F} \otimes \mathbf{F}) \cdot \text{vec}(\Sigma) + \text{vec}(\mathbf{Q}) = \mathcal{A} \text{vec}(\Sigma) + \text{vec}(\mathbf{Q}), \quad [10.2.16]$$

where

$$\mathcal{A} \equiv (\mathbf{F} \otimes \mathbf{F}). \quad [10.2.17]$$

Let $r = np$, so that $\mathbf{F}$ is an $(r \times r)$ matrix and $\mathcal{A}$ is an $(r^2 \times r^2)$ matrix. Equation [10.2.16] has the solution

$$\text{vec}(\Sigma) = [\mathbf{I}_{r^2} - \mathcal{A}]^{-1} \text{vec}(\mathbf{Q}), \quad [10.2.18]$$

provided that the matrix $[\mathbf{I}_{r^2} - \mathcal{A}]$ is nonsingular. This will be true as long as unity is not an eigenvalue of $\mathcal{A}$. But recall that the eigenvalues of $\mathbf{F} \otimes \mathbf{F}$ are all of the form $\lambda_i \lambda_j$, where λ_i and λ_j are eigenvalues of $\mathbf{F}$. Since $|\lambda_i| < 1$ for all i , it follows that all eigenvalues of $\mathcal{A}$ are inside the unit circle, meaning that $[\mathbf{I}_{r^2} - \mathcal{A}]$ is indeed nonsingular.

The first p autocovariance matrices of a $\text{VAR}(p)$ process can be calculated

10.2. Autocovariances and Convergence Results for Vector Processes 265

by substituting [10.2.12] into [10.2.18]:

$$\text{vec} \begin{bmatrix} \mathbf{\Gamma}_0 & \mathbf{\Gamma}_1 & \cdots & \mathbf{\Gamma}_{p-1} \\ \mathbf{\Gamma}'_1 & \mathbf{\Gamma}_0 & \cdots & \mathbf{\Gamma}_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{\Gamma}'_{p-1} & \mathbf{\Gamma}'_{p-2} & \cdots & \mathbf{\Gamma}_0 \end{bmatrix} = [\mathbf{I}_{r^2} - \mathcal{A}]^{-1} \text{vec}(\mathbf{Q}). \quad [10.2.19]$$

The j th autocovariance of $\boldsymbol{\xi}$ (denoted $\boldsymbol{\Sigma}_j$) can be found by postmultiplying [10.1.11] by $\boldsymbol{\xi}'_{t-j}$ and taking expectations:

$$E(\boldsymbol{\xi}_t \boldsymbol{\xi}'_{t-j}) = \mathbf{F} \cdot E(\boldsymbol{\xi}_{t-1} \boldsymbol{\xi}'_{t-j}) + E(\mathbf{v}_t \boldsymbol{\xi}'_{t-j}).$$

Thus,

$$\boldsymbol{\Sigma}_j = \mathbf{F} \boldsymbol{\Sigma}_{j-1} \quad \text{for } j = 1, 2, \dots, \quad [10.2.20]$$

or

$$\boldsymbol{\Sigma}_j = \mathbf{F}^j \boldsymbol{\Sigma} \quad \text{for } j = 1, 2, \dots. \quad [10.2.21]$$

The j th autocovariance $\mathbf{\Gamma}_j$ of the original process $\mathbf{y}_t$ is given by the first n rows and n columns of [10.2.20]:

$$\mathbf{\Gamma}_j = \boldsymbol{\Phi}_1 \mathbf{\Gamma}_{j-1} + \boldsymbol{\Phi}_2 \mathbf{\Gamma}_{j-2} + \cdots + \boldsymbol{\Phi}_p \mathbf{\Gamma}_{j-p} \quad \text{for } j = p, p+1, p+2, \dots. \quad [10.2.22]$$

10.3. The Autocovariance-Generating Function for Vector Processes

Definition of Autocovariance-Generating Function for Vector Processes

Recall that for a covariance-stationary univariate process y_t with absolutely summable autocovariances, the (scalar-valued) autocovariance-generating function $g_Y(z)$ is defined as

$$g_Y(z) \equiv \sum_{j=-\infty}^{\infty} \gamma_j z^j$$

with

$$\gamma_j \equiv E[(y_t - \mu)(y_{t-j} - \mu)]$$

and z a complex scalar. For a covariance-stationary vector process $\mathbf{y}_t$ with an absolutely summable sequence of autocovariance matrices, the analogous matrix-valued autocovariance-generating function $\mathbf{G}_Y(z)$ is defined as

$$\mathbf{G}_Y(z) \equiv \sum_{j=-\infty}^{\infty} \mathbf{\Gamma}_j z^j, \quad [10.3.1]$$

where

$$\mathbf{\Gamma}_j \equiv E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-j} - \boldsymbol{\mu})']$$

and z is again a complex scalar.

Autocovariance-Generating Function for a Vector Moving Average Process

For example, for the vector white noise process ϵ_t characterized by [10.1.5] and [10.1.6], the autocovariance-generating function is

$$\mathbf{G}_\epsilon(z) = \mathbf{\Omega}. \quad [10.3.2]$$

For the vector $MA(q)$ process of [10.2.3], the univariate expression [3.6.3] for the autocovariance-generating function generalizes to

$$\begin{aligned} \mathbf{G}_Y(z) &= (\mathbf{I}_n + \mathbf{\Theta}_1 z + \mathbf{\Theta}_2 z^2 + \cdots + \mathbf{\Theta}_q z^q) \mathbf{\Omega} \\ &\times (\mathbf{I}_n + \mathbf{\Theta}'_1 z^{-1} + \mathbf{\Theta}'_2 z^{-2} + \cdots + \mathbf{\Theta}'_q z^{-q}). \end{aligned} \quad [10.3.3]$$

This can be verified by noting that the coefficient on z^j in [10.3.3] is equal to $\mathbf{\Gamma}_j$ as given in [10.2.5].

For an $MA(\infty)$ process of the form

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{\Psi}_0 \epsilon_t + \mathbf{\Psi}_1 \epsilon_{t-1} + \mathbf{\Psi}_2 \epsilon_{t-2} + \cdots = \boldsymbol{\mu} + \mathbf{\Psi}(L) \epsilon_t,$$

with $\{\mathbf{\Psi}_k\}_{k=0}^\infty$ absolutely summable, [10.3.3] generalizes to

$$\mathbf{G}_Y(z) = [\mathbf{\Psi}(z)] \mathbf{\Omega} [\mathbf{\Psi}(z^{-1})]'. \quad [10.3.4]$$

Autocovariance-Generating Function for a Vector Autoregression

Consider the $VAR(1)$ process $\boldsymbol{\xi}_t = \mathbf{F} \boldsymbol{\xi}_{t-1} + \mathbf{v}_t$ with eigenvalues of $\mathbf{F}$ inside the unit circle and with $\boldsymbol{\xi}_t$ an $(r \times 1)$ vector and $E(\mathbf{v}_t \mathbf{v}'_t) = \mathbf{Q}$. Equation [10.3.4] implies that the autocovariance-generating function can be expressed as

$$\begin{aligned} \mathbf{G}_\xi(z) &= [\mathbf{I}_r - \mathbf{F}z]^{-1} \mathbf{Q} [\mathbf{I}_r - \mathbf{F}'z^{-1}]^{-1} \\ &= [\mathbf{I}_r + \mathbf{F}z + \mathbf{F}^2 z^2 + \mathbf{F}^3 z^3 + \cdots] \mathbf{Q} \\ &\times [\mathbf{I}_r + (\mathbf{F}')z^{-1} + (\mathbf{F}')^2 z^{-2} + (\mathbf{F}')^3 z^{-3} + \cdots]. \end{aligned} \quad [10.3.5]$$

Transformations of Vector Processes

The autocovariance-generating function of the sum of two univariate processes that are uncorrelated with each other is equal to the sum of their individual autocovariance-generating functions (equation [4.7.19]). This result readily generalizes to the vector case:

$$\begin{aligned} \mathbf{G}_{\mathbf{x}+\mathbf{w}}(z) &= \sum_{j=-\infty}^{\infty} E[(\mathbf{x}_t + \mathbf{w}_t - \boldsymbol{\mu}_x - \boldsymbol{\mu}_w) \\ &\quad \times (\mathbf{x}_{t-j} + \mathbf{w}_{t-j} - \boldsymbol{\mu}_x - \boldsymbol{\mu}_w)' z^j] \\ &= \sum_{j=-\infty}^{\infty} E[(\mathbf{x}_t - \boldsymbol{\mu}_x)(\mathbf{x}_{t-j} - \boldsymbol{\mu}_x)' z^j] \\ &\quad + \sum_{j=-\infty}^{\infty} E[(\mathbf{w}_t - \boldsymbol{\mu}_w)(\mathbf{w}_{t-j} - \boldsymbol{\mu}_w)' z^j] \\ &= \mathbf{G}_x(z) + \mathbf{G}_w(z). \end{aligned}$$

Note also that if an $(r \times 1)$ vector ξ_t is premultiplied by a nonstochastic $(n \times r)$ matrix $\mathbf{H}'$, the effect is to premultiply the autocovariance by $\mathbf{H}'$ and postmultiply by $\mathbf{H}$:

$$E[(\mathbf{H}'\xi_t - \mathbf{H}'\mu_\xi)(\mathbf{H}'\xi_{t-j} - \mathbf{H}'\mu_\xi)'] = \mathbf{H}'E[(\xi_t - \mu_\xi)(\xi_{t-j} - \mu_\xi)']\mathbf{H},$$

implying

$$\mathbf{G}_{\mathbf{H}'\xi}(z) = \mathbf{H}'\mathbf{G}_\xi(z)\mathbf{H}.$$

Putting these results together, consider ξ_t the r -dimensional $VAR(1)$ process $\xi_t = \mathbf{F}\xi_{t-1} + \mathbf{v}_t$ and a new process $\mathbf{u}_t$ given by $\mathbf{u}_t = \mathbf{H}'\xi_t + \mathbf{w}_t$ with $\mathbf{w}_t$ a white noise process that is uncorrelated with ξ_{t-j} for all j . Then

$$\mathbf{G}_\mathbf{u}(z) = \mathbf{H}'\mathbf{G}_\xi(z)\mathbf{H} + \mathbf{G}_\mathbf{w}(z),$$

or, if $\mathbf{R}$ is the variance of $\mathbf{w}_t$,

$$\mathbf{G}_\mathbf{u}(z) = \mathbf{H}'[\mathbf{I}_r - \mathbf{F}z]^{-1}\mathbf{Q}[\mathbf{I}_r - \mathbf{F}'z^{-1}]^{-1}\mathbf{H} + \mathbf{R}. \quad [10.3.6]$$

More generally, consider an $(n \times 1)$ vector $\mathbf{y}_t$ characterized by

$$\mathbf{y}_t = \mu_\mathbf{y} + \Psi(L)\epsilon_t,$$

where ϵ_t is a white noise process with variance-covariance matrix given by Ω and where $\Psi(L) = \sum_{k=0}^{\infty} \Psi_k L^k$ with $\{\Psi_k\}_{k=0}^{\infty}$ absolutely summable. Thus, the autocovariance-generating function for $\mathbf{y}$ is

$$\mathbf{G}_\mathbf{y}(z) = \Psi(z)\Omega[\Psi(z^{-1})]'. \quad [10.3.7]$$

Let $\{\mathbf{H}_k\}_{k=-\infty}^{\infty}$ be an absolutely summable sequence of $(r \times n)$ matrices, and suppose that an $(r \times 1)$ vector $\mathbf{x}_t$ is constructed from $\mathbf{y}_t$ according to

$$\mathbf{x}_t = \mathbf{H}(L)\mathbf{y}_t = \sum_{k=-\infty}^{\infty} \mathbf{H}_k \mathbf{y}_{t-k} = \mu_\mathbf{x} + \mathbf{B}(L)\epsilon_t,$$

where $\mu_\mathbf{x} = \mathbf{H}(1)\mu_\mathbf{y}$ and $\mathbf{B}(L) = \mathbf{H}(L)\Psi(L)$ as in [10.2.10] and [10.2.11]. Then the autocovariance-generating function for $\mathbf{x}$ can be found from

$$\mathbf{G}_\mathbf{x}(z) = \mathbf{B}(z)\Omega[\mathbf{B}(z^{-1})]' = [\mathbf{H}(z)\Psi(z)]\Omega[\Psi(z^{-1})]'[\mathbf{H}(z^{-1})]'. \quad [10.3.8]$$

Comparing [10.3.8] with [10.3.7], the effect of applying the filter $\mathbf{H}(L)$ to $\mathbf{y}_t$ is to premultiply the autocovariance-generating function by $\mathbf{H}(z)$ and to postmultiply by the transpose of $\mathbf{H}(z^{-1})$:

$$\mathbf{G}_\mathbf{x}(z) = [\mathbf{H}(z)]\mathbf{G}_\mathbf{y}(z)[\mathbf{H}(z^{-1})]'. \quad [10.3.9]$$

10.4. The Spectrum for Vector Processes

Let $\mathbf{y}_t$ be an $(n \times 1)$ vector with mean $E(\mathbf{y}_t) = \mu$ and k th autocovariance matrix

$$E[(\mathbf{y}_t - \mu)(\mathbf{y}_{t-k} - \mu)'] = \Gamma_k. \quad [10.4.1]$$

If $\{\Gamma_k\}_{k=-\infty}^{\infty}$ is absolutely summable and if z is a complex scalar, the autocovariance-generating function of $\mathbf{y}$ is given by

$$\mathbf{G}_\mathbf{y}(z) = \sum_{k=-\infty}^{\infty} \Gamma_k z^k. \quad [10.4.2]$$

The function $\mathbf{G}_Y(z)$ associates an $(n \times n)$ matrix of complex numbers with the complex scalar z . If [10.4.2] is divided by 2π and evaluated at $z = e^{-i\omega}$, where ω is a real scalar and $i = \sqrt{-1}$, the result is the *population spectrum* of the vector $\mathbf{y}$:

$$\mathbf{s}_Y(\omega) = (2\pi)^{-1} \mathbf{G}_Y(e^{-i\omega}) = (2\pi)^{-1} \sum_{k=-\infty}^{\infty} \mathbf{\Gamma}_k e^{-i\omega k}. \quad [10.4.3]$$

The population spectrum associates an $(n \times n)$ matrix of complex numbers with the real scalar ω .

Identical calculations to those used to establish Proposition 6.1 indicate that when any element of $\mathbf{s}_Y(\omega)$ is multiplied by $e^{i\omega k}$ and the resulting function of ω is integrated from $-\pi$ to π , the result is the corresponding element of the k th autocovariance matrix of $\mathbf{y}$:

$$\int_{-\pi}^{\pi} \mathbf{s}_Y(\omega) e^{i\omega k} d\omega = \mathbf{\Gamma}_k. \quad [10.4.4]$$

Thus, as in the univariate case, the sequence of autocovariances $\{\mathbf{\Gamma}_k\}_{k=-\infty}^{\infty}$ and the function represented by the population spectrum $\mathbf{s}_Y(\omega)$ contain the identical information.

As a special case, when $k = 0$, equation [10.4.4] implies

$$\int_{-\pi}^{\pi} \mathbf{s}_Y(\omega) d\omega = \mathbf{\Gamma}_0. \quad [10.4.5]$$

In other words, the area under the population spectrum is the unconditional variance-covariance matrix of $\mathbf{y}$.

The j th diagonal element of $\mathbf{\Gamma}_k$ is $E(y_{jt} - \mu_j)(y_{j,t-k} - \mu_j)$, the k th autocovariance of y_{jt} . Thus, the j th diagonal element of the multivariate spectrum $\mathbf{s}_Y(\omega)$ is just the univariate spectrum of the scalar y_{jt} . It follows from the properties of the univariate spectrum discussed in Chapter 6 that the diagonal elements of $\mathbf{s}_Y(\omega)$ are real-valued and nonnegative for all ω . However, the same is not true of the off-diagonal elements of $\mathbf{s}_Y(\omega)$ —in general, the off-diagonal elements of $\mathbf{s}_Y(\omega)$ will be complex numbers.

To gain further understanding of the multivariate spectrum, we concentrate on the case of $n = 2$ variables, denoted

$$\mathbf{y}_t = \begin{bmatrix} X_t \\ Y_t \end{bmatrix}.$$

The k th autocovariance matrix is then

$$\begin{aligned} \mathbf{\Gamma}_k &= E \begin{bmatrix} (X_t - \mu_X)(X_{t-k} - \mu_X) & (X_t - \mu_X)(Y_{t-k} - \mu_Y) \\ (Y_t - \mu_Y)(X_{t-k} - \mu_X) & (Y_t - \mu_Y)(Y_{t-k} - \mu_Y) \end{bmatrix} \\ &\equiv \begin{bmatrix} \gamma_{XX}^{(k)} & \gamma_{XY}^{(k)} \\ \gamma_{YX}^{(k)} & \gamma_{YY}^{(k)} \end{bmatrix}. \end{aligned} \quad [10.4.6]$$

Recall from [10.2.2] that $\mathbf{\Gamma}'_k = \mathbf{\Gamma}_{-k}$. Hence,

$$\gamma_{XX}^{(k)} = \gamma_{XX}^{(-k)} \quad [10.4.7]$$

$$\gamma_{YY}^{(k)} = \gamma_{YY}^{(-k)} \quad [10.4.8]$$

$$\gamma_{XY}^{(k)} = \gamma_{YX}^{(-k)}. \quad [10.4.9]$$

10.4. The Spectrum for Vector Processes 269

For this $n = 2$ case, the population spectrum [10.4.3] would be

$$\begin{aligned}
 s_Y(\omega) &= \frac{1}{2\pi} \begin{bmatrix} \sum_{k=-\infty}^{\infty} \gamma_{XX}^{(k)} e^{-i\omega k} & \sum_{k=-\infty}^{\infty} \gamma_{XY}^{(k)} e^{-i\omega k} \\ \sum_{k=-\infty}^{\infty} \gamma_{YX}^{(k)} e^{-i\omega k} & \sum_{k=-\infty}^{\infty} \gamma_{YY}^{(k)} e^{-i\omega k} \end{bmatrix} \\
 &= \frac{1}{2\pi} \begin{bmatrix} \sum_{k=-\infty}^{\infty} \gamma_{XX}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\} & \sum_{k=-\infty}^{\infty} \gamma_{XY}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\} \\ \sum_{k=-\infty}^{\infty} \gamma_{YX}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\} & \sum_{k=-\infty}^{\infty} \gamma_{YY}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\} \end{bmatrix}.
 \end{aligned}
 \tag{10.4.10}$$

Using [10.4.7] and [10.4.8] along with the facts that $\sin(-\omega k) = -\sin(\omega k)$ and $\sin(0) = 0$, the imaginary components disappear from the diagonal terms:

$$\begin{aligned}
 s_Y(\omega) &= \frac{1}{2\pi} \begin{bmatrix} \sum_{k=-\infty}^{\infty} \gamma_{XX}^{(k)} \cos(\omega k) & \sum_{k=-\infty}^{\infty} \gamma_{XY}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\} \\ \sum_{k=-\infty}^{\infty} \gamma_{YX}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\} & \sum_{k=-\infty}^{\infty} \gamma_{YY}^{(k)} \cos(\omega k) \end{bmatrix}.
 \end{aligned}
 \tag{10.4.11}$$

However, since in general $\gamma_{XY}^{(k)} \neq \gamma_{YX}^{(-k)}$, the off-diagonal elements are typically complex numbers.

The Cross Spectrum, Cospectrum, and Quadrature Spectrum

The lower left element of the matrix in [10.4.11] is known as the *population cross spectrum* from X to Y :

$$s_{YX}(\omega) = (2\pi)^{-1} \sum_{k=-\infty}^{\infty} \gamma_{YX}^{(k)} \{\cos(\omega k) - i \sin(\omega k)\}.
 \tag{10.4.12}$$

The cross spectrum can be written in terms of its real and imaginary components as

$$s_{YX}(\omega) = c_{YX}(\omega) + i \cdot q_{YX}(\omega).
 \tag{10.4.13}$$

The real component of the cross spectrum is known as the *cospectrum* between X and Y :

$$c_{YX}(\omega) = (2\pi)^{-1} \sum_{k=-\infty}^{\infty} \gamma_{YX}^{(k)} \cos(\omega k).
 \tag{10.4.14}$$

One can verify from [10.4.9] and the fact that $\cos(-\omega k) = \cos(\omega k)$ that

$$c_{YX}(\omega) = c_{XY}(\omega).
 \tag{10.4.15}$$

The imaginary component of the cross spectrum is known as the *quadrature spectrum* from X to Y :

$$q_{YX}(\omega) = -(2\pi)^{-1} \sum_{k=-\infty}^{\infty} \gamma_{YX}^{(k)} \sin(\omega k). \quad [10.4.16]$$

One can verify from [10.4.9] and the fact that $\sin(-\omega k) = -\sin(\omega k)$ that the quadrature spectrum from Y to X is the negative of the quadrature spectrum from X to Y :

$$q_{YX}(\omega) = -q_{XY}(\omega).$$

Recalling [10.4.13], these results imply that the off-diagonal elements of $\mathbf{s}_Y(\omega)$ are complex conjugates of each other; in general, the row j , column m element of $\mathbf{s}_Y(\omega)$ is the complex conjugate of the row m , column j element of $\mathbf{s}_Y(\omega)$.

Note that both $c_{YX}(\omega)$ and $q_{YX}(\omega)$ are real-valued periodic functions of ω :

$$\begin{aligned} c_{YX}(\omega + 2\pi j) &= c_{YX}(\omega) & \text{for } j = \pm 1, \pm 2, \dots \\ q_{YX}(\omega + 2\pi j) &= q_{YX}(\omega) & \text{for } j = \pm 1, \pm 2, \dots \end{aligned}$$

It further follows from [10.4.14] that

$$c_{YX}(-\omega) = c_{YX}(\omega),$$

while [10.4.16] implies that

$$q_{YX}(-\omega) = -q_{YX}(\omega). \quad [10.4.17]$$

Hence, the cospectrum and quadrature spectrum are fully specified by the values they assume as ω ranges between 0 and π .

Result [10.4.5] implies that the cross spectrum integrates to the unconditional covariance between X and Y :

$$\int_{-\pi}^{\pi} s_{YX}(\omega) d\omega = E(Y_t - \mu_Y)(X_t - \mu_X).$$

Observe from [10.4.17] that the quadrature spectrum integrates to zero:

$$\int_{-\pi}^{\pi} q_{YX}(\omega) d\omega = 0.$$

Hence, the covariance between X and Y can be calculated from the area under the cospectrum between X and Y :

$$\int_{-\pi}^{\pi} c_{YX}(\omega) d\omega = E(Y_t - \mu_Y)(X_t - \mu_X). \quad [10.4.18]$$

The cospectrum between X and Y at frequency ω can thus be interpreted as the portion of the covariance between X and Y that is attributable to cycles with frequency ω . Since the covariance can be positive or negative, the cospectrum can be positive or negative, and indeed, $c_{YX}(\omega)$ may be positive over some frequencies and negative over others.

The Sample Multivariate Periodogram

To gain further understanding of the cospectrum and the quadrature spectrum, let $y_1, y_2, \dots, y_T$ and $x_1, x_2, \dots, x_T$ denote samples of T observations on the two variables. If for illustration T is odd, Proposition 6.2 indicates that the value of y_t can be expressed as

$$y_t = \bar{y} + \sum_{j=1}^M \{ \hat{\alpha}_j \cdot \cos[\omega_j(t-1)] + \hat{\delta}_j \cdot \sin[\omega_j(t-1)] \}, \quad [10.4.19]$$

where $\bar{y}$ is the sample mean of y , $M = (T-1)/2$, $\omega_j = 2\pi j/T$, and

$$\hat{\alpha}_j = (2/T) \sum_{t=1}^T y_t \cdot \cos[\omega_j(t-1)] \quad [10.4.20]$$

$$\hat{\delta}_j = (2/T) \sum_{t=1}^T y_t \cdot \sin[\omega_j(t-1)]. \quad [10.4.21]$$

An analogous representation for x_t is

$$x_t = \bar{x} + \sum_{j=1}^M \{ \hat{a}_j \cdot \cos[\omega_j(t-1)] + \hat{d}_j \cdot \sin[\omega_j(t-1)] \} \quad [10.4.22]$$

$$\hat{a}_j = (2/T) \sum_{t=1}^T x_t \cdot \cos[\omega_j(t-1)] \quad [10.4.23]$$

$$\hat{d}_j = (2/T) \sum_{t=1}^T x_t \cdot \sin[\omega_j(t-1)]. \quad [10.4.24]$$

Recall from [6.2.11] that the periodic regressors in [10.4.19] all have sample mean zero and are mutually orthogonal, while

$$\sum_{t=1}^T \cos^2[\omega_j(t-1)] = \sum_{t=1}^T \sin^2[\omega_j(t-1)] = T/2. \quad [10.4.25]$$

Consider the sample covariance between x and y :

$$T^{-1} \sum_{t=1}^T (y_t - \bar{y})(x_t - \bar{x}). \quad [10.4.26]$$

Substituting [10.4.19] and [10.4.22] into [10.4.26] and exploiting the mutual orthogonality of the periodic regressors reveal that

$$\begin{aligned} & T^{-1} \sum_{t=1}^T (y_t - \bar{y})(x_t - \bar{x}) \\ &= T^{-1} \sum_{t=1}^T \left\{ \sum_{j=1}^M \{ \hat{\alpha}_j \cdot \cos[\omega_j(t-1)] + \hat{\delta}_j \cdot \sin[\omega_j(t-1)] \} \right. \\ & \quad \times \left. \sum_{j=1}^M \{ \hat{a}_j \cdot \cos[\omega_j(t-1)] + \hat{d}_j \cdot \sin[\omega_j(t-1)] \} \right\} \quad [10.4.27] \\ &= T^{-1} \sum_{t=1}^T \left\{ \sum_{j=1}^M \{ \hat{\alpha}_j \hat{a}_j \cdot \cos^2[\omega_j(t-1)] + \hat{\delta}_j \hat{d}_j \cdot \sin^2[\omega_j(t-1)] \} \right\} \\ &= (1/2) \sum_{j=1}^M (\hat{\alpha}_j \hat{a}_j + \hat{\delta}_j \hat{d}_j). \end{aligned}$$

Hence, the portion of the sample covariance between x and y that is due to their common dependence on cycles of frequency ω_j is given by

$$(1/2)(\hat{\alpha}_j \hat{a}_j + \hat{\delta}_j \hat{d}_j). \quad [10.4.28]$$

This magnitude can be related to the sample analog of the cospectrum with calculations similar to those used to establish result (c) of Proposition 6.2. Recall that since

$$\sum_{t=1}^T \cos[\omega_j(t-1)] = 0,$$

the magnitude $\hat{\alpha}_j$ in [10.4.20] can alternatively be expressed as

$$\hat{\alpha}_j = (2/T) \sum_{t=1}^T (y_t - \bar{y}) \cdot \cos[\omega_j(t-1)].$$

Thus,

$$\begin{aligned} & (\hat{a}_j + i \cdot \hat{d}_j)(\hat{\alpha}_j - i \cdot \hat{\delta}_j) \\ &= (4/T^2) \left\{ \sum_{t=1}^T (x_t - \bar{x}) \cdot \cos[\omega_j(t-1)] + i \cdot \sum_{t=1}^T (x_t - \bar{x}) \cdot \sin[\omega_j(t-1)] \right\} \\ & \quad \times \left\{ \sum_{\tau=1}^T (y_\tau - \bar{y}) \cdot \cos[\omega_j(\tau-1)] - i \cdot \sum_{\tau=1}^T (y_\tau - \bar{y}) \cdot \sin[\omega_j(\tau-1)] \right\} \\ &= (4/T^2) \left\{ \sum_{t=1}^T (x_t - \bar{x}) \cdot \exp[i \cdot \omega_j(t-1)] \right\} \left\{ \sum_{\tau=1}^T (y_\tau - \bar{y}) \cdot \exp[-i \cdot \omega_j(\tau-1)] \right\} \\ &= (4/T^2) \left\{ \sum_{t=1}^T (x_t - \bar{x})(y_t - \bar{y}) + \sum_{t=1}^{T-1} (x_t - \bar{x})(y_{t+1} - \bar{y}) \cdot \exp[-i\omega_j] \right. \\ & \quad + \sum_{t=2}^T (x_t - \bar{x})(y_{t-1} - \bar{y}) \cdot \exp[i\omega_j] + \sum_{t=1}^{T-2} (x_t - \bar{x})(y_{t+2} - \bar{y}) \cdot \exp[-2i\omega_j] \\ & \quad + \sum_{t=3}^T (x_t - \bar{x})(y_{t-2} - \bar{y}) \cdot \exp[2i\omega_j] + \cdots + (x_1 - \bar{x})(y_T - \bar{y}) \cdot \exp[-(T-1)i\omega_j] \\ & \quad \left. + (x_T - \bar{x})(y_1 - \bar{y}) \cdot \exp[(T-1)i\omega_j] \right\} \\ &= (4/T) \left\{ \hat{\gamma}_{yx}^{(0)} + \hat{\gamma}_{yx}^{(1)} \cdot \exp[-i\omega_j] + \hat{\gamma}_{yx}^{(-1)} \cdot \exp[i\omega_j] \right. \\ & \quad + \hat{\gamma}_{yx}^{(2)} \cdot \exp[-2i\omega_j] + \hat{\gamma}_{yx}^{(-2)} \cdot \exp[2i\omega_j] + \cdots \\ & \quad \left. + \hat{\gamma}_{yx}^{(T-1)} \cdot \exp[-(T-1)i\omega_j] + \hat{\gamma}_{yx}^{(-T+1)} \cdot \exp[(T-1)i\omega_j] \right\}, \quad [10.4.29] \end{aligned}$$

where $\hat{\gamma}_{yx}^{(k)}$ is the sample covariance between the value of y and the value that x assumed k periods earlier:

$$\hat{\gamma}_{yx}^{(k)} = \begin{cases} (1/T) \sum_{t=1}^{T-k} (x_t - \bar{x})(y_{t+k} - \bar{y}) & \text{for } k = 0, 1, 2, \dots, T-1 \\ (1/T) \sum_{t=-k+1}^T (x_t - \bar{x})(y_{t+k} - \bar{y}) & \text{for } k = -1, -2, \dots, -T+1. \end{cases} \quad [10.4.30]$$

Result [10.4.29] implies that

$$\begin{aligned} \frac{1}{2}(\hat{a}_j + i \cdot \hat{d}_j)(\hat{\alpha}_j - i \cdot \hat{\delta}_j) &= (2/T) \sum_{k=-T+1}^{T-1} \hat{\gamma}_{yx}^{(k)} \cdot \exp[-ki\omega_j] \\ &= (4\pi/T) \cdot \hat{s}_{yx}(\omega_j), \end{aligned} \quad [10.4.31]$$

where $\hat{s}_{yx}(\omega_j)$ is the *sample cross periodogram* from x to y at frequency ω_j , or the lower left element of the *sample multivariate periodogram*:

$$\hat{s}_y(\omega) = (2\pi)^{-1} \begin{bmatrix} \sum_{k=-T+1}^{T-1} \hat{\gamma}_{xx}^{(k)} e^{-i\omega k} & \sum_{k=-T+1}^{T-1} \hat{\gamma}_{xy}^{(k)} e^{-i\omega k} \\ \sum_{k=-T+1}^{T-1} \hat{\gamma}_{yx}^{(k)} e^{-i\omega k} & \sum_{k=-T+1}^{T-1} \hat{\gamma}_{yy}^{(k)} e^{-i\omega k} \end{bmatrix} = \begin{bmatrix} \hat{s}_{xx}(\omega) & \hat{s}_{xy}(\omega) \\ \hat{s}_{yx}(\omega) & \hat{s}_{yy}(\omega) \end{bmatrix}.$$

Expression [10.4.31] states that the sample cross periodogram from x to y at frequency ω_j can be expressed as

$$\begin{aligned} \hat{s}_{yx}(\omega_j) &= [T/(8\pi)] \cdot (\hat{a}_j + i \cdot \hat{d}_j)(\hat{\alpha}_j - i \cdot \hat{\delta}_j) \\ &= [T/(8\pi)] \cdot (\hat{a}_j \hat{\alpha}_j + \hat{d}_j \hat{\delta}_j) + i \cdot [T/(8\pi)] \cdot (\hat{d}_j \hat{\alpha}_j - \hat{a}_j \hat{\delta}_j). \end{aligned}$$

The real component is the sample analog of the cospectrum, while the imaginary component is the sample analog of the quadrature spectrum:

$$\hat{s}_{yx}(\omega_j) = \hat{c}_{yx}(\omega_j) + i \cdot \hat{q}_{yx}(\omega_j), \quad [10.4.32]$$

where

$$\hat{c}_{yx}(\omega_j) = [T/(8\pi)] \cdot (\hat{a}_j \hat{\alpha}_j + \hat{d}_j \hat{\delta}_j) \quad [10.4.33]$$

$$\hat{q}_{yx}(\omega_j) = [T/(8\pi)] \cdot (\hat{d}_j \hat{\alpha}_j - \hat{a}_j \hat{\delta}_j). \quad [10.4.34]$$

Comparing [10.4.33] with [10.4.28], the sample cospectrum evaluated at ω_j is proportional to the portion of the sample covariance between y and x that is attributable to cycles with frequency ω_j . The population cospectrum admits an analogous interpretation as the portion of the population covariance between Y and X attributable to cycles with frequency ω based on a multivariate version of the spectral representation theorem.

What interpretation are we to attach to the quadrature spectrum? Consider using the weights in [10.4.22] to construct a new series x_t^* by shifting the phase of each of the periodic functions by a quarter cycle:

$$\begin{aligned} x_t^* &= \bar{x} + \sum_{j=1}^M \{ \hat{a}_j \cdot \cos[\omega_j(t-1) + (\pi/2)] \\ &\quad + \hat{d}_j \cdot \sin[\omega_j(t-1) + (\pi/2)] \}. \end{aligned} \quad [10.4.35]$$

The variable x_t^* is driven by the same cycles as x_t , except that at date $t = 1$ each cycle is one-quarter of the way through rather than just beginning as in the case of x_t .

Since $\sin[\theta + (\pi/2)] = \cos(\theta)$ and since $\cos[\theta + (\pi/2)] = -\sin(\theta)$, the variable x_t^* can alternatively be described as

$$x_t^* = \bar{x} + \sum_{j=1}^M \{ \hat{d}_j \cdot \cos[\omega_j(t-1)] - \hat{a}_j \cdot \sin[\omega_j(t-1)] \}. \quad [10.4.36]$$

As in [10.4.27], the sample covariance between y_t and x_t^* is found to be

$$T^{-1} \sum_{t=1}^T (y_t - \bar{y})(x_t^* - \bar{x}) = (1/2) \sum_{j=1}^M (\hat{\alpha}_j \hat{d}_j - \hat{\delta}_j \hat{a}_j).$$

Comparing this with [10.4.34], the sample quadrature spectrum from x to y at frequency ω_j is proportional to the portion of the sample covariance between x^* and y that is due to cycles of frequency ω_j . Cycles of frequency ω_j may be quite important for both x and y individually (as reflected by large values for $\hat{s}_{xx}(\omega)$ and $\hat{s}_{yy}(\omega)$) yet fail to produce much contemporaneous covariance between the variables because at any given date the two series are in a different phase of the cycle. For example, the variable x may respond to an economic recession sooner than y . The quadrature spectrum looks for evidence of such out-of-phase cycles.

Coherence, Phase, and Gain

The *population coherence* between X and Y is a measure of the degree to which X and Y are jointly influenced by cycles of frequency ω . This measure combines the inferences of the cospectrum and the quadrature spectrum, and is defined as¹

$$h_{YX}(\omega) = \frac{[c_{YX}(\omega)]^2 + [q_{YX}(\omega)]^2}{s_{YY}(\omega)s_{XX}(\omega)},$$

assuming that $s_{YY}(\omega)$ and $s_{XX}(\omega)$ are nonzero. If $s_{YY}(\omega)$ or $s_{XX}(\omega)$ is zero, the coherence is defined to be zero. It can be shown that $0 \leq h_{YX}(\omega) \leq 1$ for all ω as long as X and Y are covariance-stationary with absolutely summable autocovariance matrices.² If $h_{YX}(\omega)$ is large, this indicates that Y and X have important cycles of frequency ω in common.

The cospectrum and quadrature spectrum can alternatively be described in polar coordinate form. In this notation, the population cross spectrum from X to Y is written as

$$s_{YX}(\omega) = c_{YX}(\omega) + i \cdot q_{YX}(\omega) = R(\omega) \cdot \exp[i \cdot \theta(\omega)], \quad [10.4.37]$$

where

$$R(\omega) = \{[c_{YX}(\omega)]^2 + [q_{YX}(\omega)]^2\}^{1/2} \quad [10.4.38]$$

and $\theta(\omega)$ represents the radian angle satisfying

$$\frac{\sin[\theta(\omega)]}{\cos[\theta(\omega)]} = \frac{q_{YX}(\omega)}{c_{YX}(\omega)}. \quad [10.4.39]$$

The function $R(\omega)$ is sometimes described as the *gain* while $\theta(\omega)$ is called the *phase*.³

¹The coherence is sometimes alternatively defined as the square root of this magnitude. The sample coherence based on the unsmoothed periodogram is identically equal to 1.

²See, for example, Fuller (1976, p. 156).

³The gain is sometimes alternatively defined as $R(\omega)/s_{XX}(\omega)$.

The Population Spectrum for Vector MA and AR Processes

Let $\mathbf{y}_t$ be a vector $MA(\infty)$ process with absolutely summable moving average coefficients:

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\Psi}(L)\boldsymbol{\varepsilon}_t,$$

where

$$E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_\tau') = \begin{cases} \boldsymbol{\Omega} & \text{for } t = \tau \\ \mathbf{0} & \text{otherwise.} \end{cases}$$

Substituting [10.3.4] into [10.4.3] reveals that the population spectrum for $\mathbf{y}_t$ can be calculated as

$$\mathbf{s}_Y(\omega) = (2\pi)^{-1} [\boldsymbol{\Psi}(e^{-i\omega})] \boldsymbol{\Omega} [\boldsymbol{\Psi}(e^{i\omega})]'. \quad [10.4.40]$$

For example, the population spectrum for a stationary $VAR(p)$ as written in [10.1.4] is

$$\begin{aligned} \mathbf{s}_Y(\omega) = (2\pi)^{-1} \{ \mathbf{I}_n - \boldsymbol{\Phi}_1 e^{-i\omega} - \boldsymbol{\Phi}_2 e^{-2i\omega} - \dots - \boldsymbol{\Phi}_p e^{-pi\omega} \}^{-1} \boldsymbol{\Omega} \\ \times \{ \mathbf{I}_n - \boldsymbol{\Phi}_1' e^{i\omega} - \boldsymbol{\Phi}_2' e^{2i\omega} - \dots - \boldsymbol{\Phi}_p' e^{pi\omega} \}^{-1}. \end{aligned} \quad [10.4.41]$$

Estimating the Population Spectrum

If an observed time series $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T$ can be reasonably described by a p th-order vector autoregression, one good approach to estimating the population spectrum is to estimate the parameters of the vector autoregression [10.1.4] by *OLS* and then substitute these parameter estimates into equation [10.4.41].

Alternatively, the sample cross periodogram from x to y at frequency $\omega_j = 2\pi j/T$ can be calculated from [10.4.32] to [10.4.34], where $\hat{\alpha}_j$, $\hat{\delta}_j$, $\hat{a}_j$, and $\hat{d}_j$ are as defined in [10.4.20] through [10.4.24]. One would want to smooth these to obtain a more useful estimate of the population cross spectrum. For example, one reasonable estimate of the population cospectrum between X and Y at frequency ω_j would be

$$\hat{c}_{YX}(\omega_j) = \sum_{m=-h}^h \left\{ \frac{h+1-|m|}{(h+1)^2} \right\} \hat{c}_{YX}(\omega_{j+m}),$$

where $\hat{c}_{YX}(\omega_{j+m})$ denotes the estimate in [10.4.33] evaluated at frequency $\omega_{j+m} = 2\pi(j+m)/T$ and h is a bandwidth parameter reflecting how many different frequencies are to be used in estimating the cospectrum at frequency ω_j .

Another approach is to express the smoothing in terms of weighting coefficients κ_k^* to be applied to $\hat{\Gamma}_k$ when the population autocovariances in expression [10.4.3] are replaced by sample autocovariances. Such an estimate would take the form

$$\hat{\mathbf{s}}_Y(\omega) = (2\pi)^{-1} \left\{ \hat{\Gamma}_0 + \sum_{k=1}^{T-1} \kappa_k^* [\hat{\Gamma}_k e^{-i\omega k} + \hat{\Gamma}_k' e^{i\omega k}] \right\}$$

where

$$\hat{\mathbf{r}}_k = T^{-1} \sum_{t=k+1}^T (\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{y}_{t-k} - \bar{\mathbf{y}})'$$

$$\bar{\mathbf{y}} = T^{-1} \sum_{t=1}^T \mathbf{y}_t.$$

For example, the modified Bartlett estimate of the multivariate spectrum is

$$\hat{\mathbf{s}}_{\mathbf{y}}(\omega) = (2\pi)^{-1} \left\{ \hat{\mathbf{r}}_0 + \sum_{k=1}^q \left[1 - \frac{k}{q+1} \right] [\hat{\mathbf{r}}_k e^{-i\omega k} + \hat{\mathbf{r}}_k' e^{i\omega k}] \right\}. \quad [10.4.42]$$

Filters

Let $\mathbf{x}_t$ be an r -dimensional covariance-stationary process with absolutely summable autocovariances and with $(r \times r)$ population spectrum denoted $\mathbf{s}_{\mathbf{x}}(\omega)$. Let $\{\mathbf{H}_k\}_{k=-\infty}^{\infty}$ be an absolutely summable sequence of $(n \times r)$ matrices, and let $\mathbf{y}_t$ denote the n -dimensional vector process given by

$$\mathbf{y}_t = \mathbf{H}(L)\mathbf{x}_t = \sum_{k=-\infty}^{\infty} \mathbf{H}_k \mathbf{x}_{t-k}.$$

It follows from [10.3.9] that the population spectrum of $\mathbf{y}$ (denoted $\mathbf{s}_{\mathbf{y}}(\omega)$) is related to that of $\mathbf{x}$ according to

$$\underset{(n \times n)}{\mathbf{s}_{\mathbf{y}}(\omega)} = \underset{(n \times r)}{[\mathbf{H}(e^{-i\omega})]} \underset{(r \times r)}{\mathbf{s}_{\mathbf{x}}(\omega)} \underset{(r \times n)}{[\mathbf{H}(e^{i\omega})]}' . \quad [10.4.43]$$

As a special case of this result, let X_t be a univariate stationary stochastic process with continuous spectrum $s_X(\omega)$, and let u_t be a second univariate stationary stochastic process with continuous spectrum $s_U(\omega)$, where X_t and u_τ are uncorrelated for all t and τ . Thus, the population spectrum of the vector $\mathbf{x}_t \equiv (X_t, u_t)'$ is given by

$$\mathbf{s}_{\mathbf{x}}(\omega) = \begin{bmatrix} s_{XX}(\omega) & 0 \\ 0 & s_{UU}(\omega) \end{bmatrix}.$$

Define a new series Y_t according to

$$Y_t = \sum_{k=-\infty}^{\infty} h_k X_{t-k} + u_t \equiv h(L)X_t + u_t, \quad [10.4.44]$$

where $\{h_k\}_{k=-\infty}^{\infty}$ is absolutely summable. Note that the vector $\mathbf{y}_t \equiv (X_t, Y_t)'$ is obtained from the original vector $\mathbf{x}_t$ by the filter

$$\mathbf{y}_t = \mathbf{H}(L)\mathbf{x}_t,$$

where

$$\mathbf{H}(L) = \begin{bmatrix} 1 & 0 \\ h(L) & 1 \end{bmatrix}.$$

It follows from [10.4.43] that the spectrum of y is given by

$$\begin{aligned} \mathbf{s}_Y(\omega) &= \begin{bmatrix} 1 & 0 \\ h(e^{-i\omega}) & 1 \end{bmatrix} \begin{bmatrix} s_{XX}(\omega) & 0 \\ 0 & s_{UU}(\omega) \end{bmatrix} \begin{bmatrix} 1 & h(e^{i\omega}) \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} s_{XX}(\omega) & s_{XX}(\omega)h(e^{i\omega}) \\ h(e^{-i\omega})s_{XX}(\omega) & h(e^{-i\omega})s_{XX}(\omega)h(e^{i\omega}) + s_{UU}(\omega) \end{bmatrix}, \end{aligned} \quad [10.4.45]$$

where

$$h(e^{-i\omega}) = \sum_{k=-\infty}^{\infty} h_k e^{-i\omega k}. \quad [10.4.46]$$

The lower left element of the matrix in [10.4.45] indicates that when Y_t and X_t are related according to [10.4.44], the cross spectrum from X to Y can be calculated by multiplying [10.4.46] by the spectrum of X .

We can also imagine going through these steps in reverse order. Specifically, suppose we are given an observed vector $\mathbf{y}_t = (X_t, Y_t)'$ with absolutely summable autocovariance matrices and with population spectrum given by

$$\mathbf{s}_Y(\omega) = \begin{bmatrix} s_{XX}(\omega) & s_{XY}(\omega) \\ s_{YX}(\omega) & s_{YY}(\omega) \end{bmatrix}. \quad [10.4.47]$$

Then the linear projection of Y_t on $\{X_{t-k}\}_{k=-\infty}^{\infty}$ exists and is of the form of [10.4.44], where u_t would now be regarded as the population residual associated with the linear projection. The sequence of linear projection coefficients $\{h_k\}_{k=-\infty}^{\infty}$ can be summarized in terms of the function of ω given in [10.4.46]. Comparing the lower left elements of [10.4.47] and [10.4.45], this function must satisfy

$$h(e^{-i\omega})s_{XX}(\omega) = s_{YX}(\omega).$$

In other words, the function $h(e^{-i\omega})$ can be calculated from

$$h(e^{-i\omega}) = \frac{s_{YX}(\omega)}{s_{XX}(\omega)}, \quad [10.4.48]$$

assuming that $s_{XX}(\omega)$ is not zero. When $s_{XX}(\omega) = 0$, we set $h(e^{-i\omega}) = 0$. This magnitude, the ratio of the cross spectrum from X to Y to the spectrum of X , is known as the *transfer function* from X to Y .

The principles underlying [10.4.4] can further be used to uncover individual transfer function coefficients:

$$h_k = (2\pi)^{-1} \int_{-\pi}^{\pi} h(e^{-i\omega}) e^{i\omega k} d\omega.$$

In other words, given an observed vector $(X_t, Y_t)'$ with absolutely summable autocovariance matrices and thus with continuous population spectrum of the form of [10.4.47], the coefficient on X_{t-k} in the population linear projection of Y_t on $\{X_{t-k}\}_{k=-\infty}^{\infty}$ can be calculated from

$$h_k = (2\pi)^{-1} \int_{-\pi}^{\pi} \frac{s_{YX}(\omega)}{s_{XX}(\omega)} e^{i\omega k} d\omega. \quad [10.4.49]$$

10.5. The Sample Mean of a Vector Process

Variance of the Sample Mean

Suppose we have a sample of size T , $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T\}$, drawn from an n -dimensional covariance-stationary process with

$$E(\mathbf{y}_t) = \boldsymbol{\mu} \quad [10.5.1]$$

$$E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-j} - \boldsymbol{\mu})'] = \boldsymbol{\Gamma}_j. \quad [10.5.2]$$

Consider the properties of the sample mean,

$$\bar{\mathbf{y}}_T = (1/T) \sum_{t=1}^T \mathbf{y}_t. \quad [10.5.3]$$

As in the discussion in Section 7.2 of the sample mean of a scalar process, it is clear that $E(\bar{\mathbf{y}}_T) = \boldsymbol{\mu}$ and

$$\begin{aligned} E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})'] &= (1/T^2)E\{(\mathbf{y}_1 - \boldsymbol{\mu})[(\mathbf{y}_1 - \boldsymbol{\mu})' + (\mathbf{y}_2 - \boldsymbol{\mu})' + \dots + (\mathbf{y}_T - \boldsymbol{\mu})'] \\ &\quad + (\mathbf{y}_2 - \boldsymbol{\mu})[(\mathbf{y}_1 - \boldsymbol{\mu})' + (\mathbf{y}_2 - \boldsymbol{\mu})' + \dots + (\mathbf{y}_T - \boldsymbol{\mu})'] \\ &\quad + (\mathbf{y}_3 - \boldsymbol{\mu})[(\mathbf{y}_1 - \boldsymbol{\mu})' + (\mathbf{y}_2 - \boldsymbol{\mu})' + \dots + (\mathbf{y}_T - \boldsymbol{\mu})'] \\ &\quad + \dots + (\mathbf{y}_T - \boldsymbol{\mu})[(\mathbf{y}_1 - \boldsymbol{\mu})' + (\mathbf{y}_2 - \boldsymbol{\mu})' + \dots + (\mathbf{y}_T - \boldsymbol{\mu})']\} \\ &= (1/T^2)\{\boldsymbol{\Gamma}_0 + \boldsymbol{\Gamma}_{-1} + \dots + \boldsymbol{\Gamma}_{-(T-1)}\} \quad [10.5.4] \\ &\quad + \{\boldsymbol{\Gamma}_1 + \boldsymbol{\Gamma}_0 + \boldsymbol{\Gamma}_{-1} + \dots + \boldsymbol{\Gamma}_{-(T-2)}\} \\ &\quad + \{\boldsymbol{\Gamma}_2 + \boldsymbol{\Gamma}_1 + \boldsymbol{\Gamma}_0 + \boldsymbol{\Gamma}_{-1} + \dots + \boldsymbol{\Gamma}_{-(T-3)}\} \\ &\quad + \dots + \{\boldsymbol{\Gamma}_{T-1} + \boldsymbol{\Gamma}_{T-2} + \boldsymbol{\Gamma}_{T-3} + \dots + \boldsymbol{\Gamma}_0\} \\ &= (1/T^2)\{T\boldsymbol{\Gamma}_0 + (T-1)\boldsymbol{\Gamma}_1 + (T-2)\boldsymbol{\Gamma}_2 + \dots + \boldsymbol{\Gamma}_{T-1} \\ &\quad + (T-1)\boldsymbol{\Gamma}_{-1} + (T-2)\boldsymbol{\Gamma}_{-2} + \dots + \boldsymbol{\Gamma}_{-(T-1)}\}. \end{aligned}$$

Thus,

$$\begin{aligned} T \cdot E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})'] &= \boldsymbol{\Gamma}_0 + [(T-1)/T]\boldsymbol{\Gamma}_1 + [(T-2)/T]\boldsymbol{\Gamma}_2 + \dots \\ &\quad + [1/T]\boldsymbol{\Gamma}_{T-1} + [(T-1)/T]\boldsymbol{\Gamma}_{-1} + [(T-2)/T]\boldsymbol{\Gamma}_{-2} \\ &\quad + \dots + [1/T]\boldsymbol{\Gamma}_{-(T-1)}. \end{aligned} \quad [10.5.5]$$

As in the univariate case, the weights on $\boldsymbol{\Gamma}_k$ for $|k|$ small go to unity as $T \rightarrow \infty$, and higher autocovariances go to zero for a covariance-stationary process. Hence, we have the following generalization of Proposition 7.5.

Proposition 10.5: *Let $\mathbf{y}_t$ be a covariance-stationary process with moments given by [10.5.1] and [10.5.2] and with absolutely summable autocovariances. Then the sample mean [10.5.3] satisfies*

$$(a) \quad \bar{\mathbf{y}}_T \xrightarrow{p} \boldsymbol{\mu}$$

$$(b) \quad \lim_{T \rightarrow \infty} \{T \cdot E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})']\} = \sum_{v=-\infty}^{\infty} \boldsymbol{\Gamma}_v.$$

The proof of Proposition 10.5 is virtually identical to that of Proposition 7.5. Consider the following $(n \times n)$ matrix:

$$\sum_{v=-\infty}^{\infty} \Gamma_v - T \cdot E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})'] = \sum_{|v| \geq T} \Gamma_v + \sum_{v=-(T-1)}^{T-1} (|v|/T) \Gamma_v, \quad [10.5.6]$$

where the equality follows from [10.5.5]. Let $\gamma_{ij}^{(v)}$ denote the row i , column j element of Γ_v . The row i , column j element of the matrix in [10.5.6] can then be written

$$\sum_{|v| \geq T} \gamma_{ij}^{(v)} + \sum_{v=-(T-1)}^{T-1} (|v|/T) \gamma_{ij}^{(v)}.$$

Absolutely summability of $\{\Gamma_v\}_{v=-\infty}^{\infty}$ implies that for any $\varepsilon > 0$ there exists a q such that

$$\sum_{|v| > q} |\gamma_{ij}^{(v)}| < \varepsilon/2.$$

Thus,

$$\left| \sum_{|v| \geq T} \gamma_{ij}^{(v)} + \sum_{v=-(T-1)}^{T-1} (|v|/T) \gamma_{ij}^{(v)} \right| < \varepsilon/2 + \sum_{v=-q}^q (|v|/T) |\gamma_{ij}^{(v)}|.$$

This sum can be made less than ε by choosing T sufficiently large. This establishes claim (b) of Proposition 10.5. From this result, $E(\bar{y}_{i,T} - \mu_i)^2 \rightarrow 0$ for each i , implying that $\bar{y}_{i,T} \xrightarrow{p} \mu_i$.

Consistent Estimation of T Times the Variance of the Sample Mean

Hypothesis tests about the sample mean require an estimate of the matrix in result (b) of Proposition 10.5. Let $\mathbf{S}$ represent this matrix:

$$\mathbf{S} \equiv \lim_{T \rightarrow \infty} T \cdot E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})']. \quad [10.5.7]$$

If the data were generated by a vector $MA(q)$ process, then result (b) would imply

$$\mathbf{S} = \sum_{v=-q}^q \Gamma_v. \quad [10.5.8]$$

A natural estimate then is

$$\hat{\mathbf{S}} = \hat{\Gamma}_0 + \sum_{v=1}^q (\hat{\Gamma}_v + \hat{\Gamma}_v'), \quad [10.5.9]$$

where

$$\hat{\Gamma}_v = (1/T) \sum_{t=v+1}^T (\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{y}_{t-v} - \bar{\mathbf{y}})'. \quad [10.5.10]$$

As long as $\mathbf{y}_t$ is ergodic for second moments, [10.5.9] gives a consistent estimate of [10.5.8]. Indeed, Hansen (1982) and White (1984, Chapter 6) noted that [10.5.9] gives a consistent estimate of the asymptotic variance of the sample mean for a broad class of processes exhibiting time-dependent heteroskedasticity and autocorrelation. To see why, note that for a process satisfying $E(\mathbf{y}_t) = \boldsymbol{\mu}$ with

time-varying second moments, the variance of the sample mean is given by

$$\begin{aligned} & E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})'] \\ &= E\left[\left(\frac{1}{T} \sum_{t=1}^T (\mathbf{y}_t - \boldsymbol{\mu})\right) \left[\left(\frac{1}{T} \sum_{s=1}^T (\mathbf{y}_s - \boldsymbol{\mu})\right)'\right]\right] \quad [10.5.11] \\ &= (1/T^2) \sum_{t=1}^T \sum_{s=1}^T E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_s - \boldsymbol{\mu})']. \end{aligned}$$

Suppose, first, that $E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_s - \boldsymbol{\mu})'] = \mathbf{0}$ for $|t - s| > q$, as was the case for the vector $MA(q)$ process, though we generalize from the $MA(q)$ process to allow $E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_s - \boldsymbol{\mu})']$ to be a function of t for $|t - s| \leq q$. Then [10.5.11] implies

$$\begin{aligned} & T \cdot E[(\bar{\mathbf{y}}_T - \boldsymbol{\mu})(\bar{\mathbf{y}}_T - \boldsymbol{\mu})'] \\ &= (1/T) \sum_{t=1}^T E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_t - \boldsymbol{\mu})'] \\ &+ (1/T) \sum_{t=2}^T \{E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-1} - \boldsymbol{\mu})'] + E[(\mathbf{y}_{t-1} - \boldsymbol{\mu})(\mathbf{y}_t - \boldsymbol{\mu})']\} \\ &+ (1/T) \sum_{t=3}^T \{E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-2} - \boldsymbol{\mu})'] + E[(\mathbf{y}_{t-2} - \boldsymbol{\mu})(\mathbf{y}_t - \boldsymbol{\mu})']\} + \cdots \\ &+ (1/T) \sum_{t=q+1}^T \{E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-q} - \boldsymbol{\mu})'] + E[(\mathbf{y}_{t-q} - \boldsymbol{\mu})(\mathbf{y}_t - \boldsymbol{\mu})']\}. \quad [10.5.12] \end{aligned}$$

The estimate [10.5.9] replaces

$$(1/T) \sum_{t=v+1}^T E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_{t-v} - \boldsymbol{\mu})'] \quad [10.5.13]$$

in [10.5.12] with

$$(1/T) \sum_{t=v+1}^T (\mathbf{y}_t - \bar{\mathbf{y}}_T)(\mathbf{y}_{t-v} - \bar{\mathbf{y}}_T)', \quad [10.5.14]$$

and thus [10.5.9] provides a consistent estimate of the limit of [10.5.12] whenever [10.5.14] converges in probability to [10.5.13]. Hence, the estimator proposed in [10.5.9] can give a consistent estimate of T times the variance of the sample mean in the presence of both heteroskedasticity and autocorrelation up through order q .

More generally, even if $E[(\mathbf{y}_t - \boldsymbol{\mu})(\mathbf{y}_s - \boldsymbol{\mu})']$ is nonzero for all t and s , as long as this matrix goes to zero sufficiently quickly as $|t - s| \rightarrow \infty$, then there is still a sense in which $\hat{\mathbf{S}}_T$ in [10.5.9] can provide a consistent estimate of $\mathbf{S}$. Specifically, if, as the sample size T grows, a larger number of sample autocovariances q is used to form the estimate, then $\hat{\mathbf{S}}_T \xrightarrow{p} \mathbf{S}$ (see White, 1984, p. 155).

The Newey-West Estimator

Although [10.5.9] gives a consistent estimate of $\mathbf{S}$, it has the drawback that [10.5.9] need not be positive semidefinite in small samples. If $\hat{\mathbf{S}}$ is not positive semidefinite, then some linear combination of the elements of $\bar{\mathbf{y}}$ is asserted to have a negative variance, a considerable handicap in forming a hypothesis test!

Newey and West (1987) suggested the alternative estimate

$$\tilde{\mathbf{S}} = \hat{\mathbf{r}}_0 + \sum_{v=1}^q \left[1 - \frac{v}{q+1}\right] (\hat{\mathbf{r}}_v + \hat{\mathbf{r}}_v'), \quad [10.5.15]$$

where $\hat{\Gamma}_v$ is given by [10.5.10]. For example, for $q = 2$,

$$\hat{\mathbf{S}} = \hat{\Gamma}_0 + \frac{2}{3}(\hat{\Gamma}_1 + \hat{\Gamma}_1') + \frac{1}{3}(\hat{\Gamma}_2 + \hat{\Gamma}_2').$$

Newey and West showed that $\hat{\mathbf{S}}$ is positive semidefinite by construction and has the same consistency properties that were noted for $\hat{\mathbf{S}}$, namely, that if q and T both go to infinity with $q/T^{1/4} \rightarrow 0$, then $\hat{\mathbf{S}}_T \xrightarrow{p} \mathbf{S}$.

Application: Autocorrelation- and Heteroskedasticity-Consistent Standard Errors for Linear Regressions

As an application of using the Newey-West weighting, consider the linear regression model

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + u_t$$

for $\mathbf{x}_t$ a $(k \times 1)$ vector of explanatory variables. Recall from equation [8.2.6] that the deviation of the *OLS* estimate $\mathbf{b}_T$ from the true value $\boldsymbol{\beta}$ satisfies

$$\sqrt{T}(\mathbf{b}_T - \boldsymbol{\beta}) = \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \left[(1/\sqrt{T}) \sum_{t=1}^T \mathbf{x}_t u_t \right]. \quad [10.5.16]$$

In calculating the asymptotic distribution of the *OLS* estimate $\mathbf{b}_T$, we usually assume that the first term in [10.5.16] converges in probability to $\mathbf{Q}^{-1}$:

$$\left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right]^{-1} \xrightarrow{p} \mathbf{Q}^{-1}. \quad [10.5.17]$$

The second term in [10.5.16] can be viewed as $\sqrt{T}$ times the sample mean of the $(k \times 1)$ vector $\mathbf{x}_t u_t$:

$$\begin{aligned} \left[(1/\sqrt{T}) \sum_{t=1}^T \mathbf{x}_t u_t \right] &= (\sqrt{T})(1/T) \sum_{t=1}^T \mathbf{y}_t, \\ &= \sqrt{T} \cdot \bar{\mathbf{y}}_T, \end{aligned} \quad [10.5.18]$$

where $\mathbf{y}_t \equiv \mathbf{x}_t u_t$. Provided that $E(u_t | \mathbf{x}_t) = 0$, the vector $\mathbf{y}_t$ has mean zero. We can allow for conditional heteroskedasticity, autocorrelation, and time variation in the second moments of $\mathbf{y}_t$, as long as

$$\mathbf{S} \equiv \lim_{T \rightarrow \infty} T \cdot E(\bar{\mathbf{y}}_T \bar{\mathbf{y}}_T')$$

exists. Under general conditions,⁴ it then turns out that

$$\left[(1/\sqrt{T}) \sum_{t=1}^T \mathbf{x}_t u_t \right] = \sqrt{T} \cdot \bar{\mathbf{y}}_T \xrightarrow{L} N(\mathbf{0}, \mathbf{S}).$$

Substituting this and [10.5.17] into [10.5.16],

$$\sqrt{T}(\mathbf{b}_T - \boldsymbol{\beta}) \xrightarrow{L} N(\mathbf{0}, \mathbf{Q}^{-1} \mathbf{S} \mathbf{Q}^{-1}). \quad [10.5.19]$$

In light of the foregoing discussion, we might hope to estimate $\mathbf{S}$ by

$$\hat{\mathbf{S}}_T = \hat{\Gamma}_{0,T} + \sum_{v=1}^q \left[1 - \frac{v}{q+1} \right] (\hat{\Gamma}_{v,T} + \hat{\Gamma}_{v,T}'). \quad [10.5.20]$$

⁴See, for example, White (1984, p. 119).

Here,

$$\hat{\mathbf{r}}_{v,T} = (1/T) \sum_{t=v+1}^T (\mathbf{x}_t \hat{u}_{t,T} \hat{u}_{t-v,T} \mathbf{x}'_{t-v}),$$

$\hat{u}_{t,T}$ is the *OLS* residual for date t in a sample of size T ($\hat{u}_{t,T} = y_t - \mathbf{x}'_t \mathbf{b}_T$), and q is a lag length beyond which we are willing to assume that the correlation between $\mathbf{x}_t \hat{u}_t$ and $\mathbf{x}_{t-v} \hat{u}_{t-v}$ is essentially zero. Clearly, $\mathbf{Q}$ is consistently estimated by $\hat{\mathbf{Q}}_T = (1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t$. Substituting $\hat{\mathbf{Q}}_T$ and $\hat{\mathbf{S}}_T$ into [10.5.19], the suggestion is to treat the *OLS* estimate $\mathbf{b}_T$ as if

$$\mathbf{b}_T \approx N(\boldsymbol{\beta}, (\hat{\mathbf{V}}_T/T))$$

where

$$\begin{aligned} \hat{\mathbf{V}}_T &= \hat{\mathbf{Q}}_T^{-1} \hat{\mathbf{S}}_T \hat{\mathbf{Q}}_T^{-1} \\ &= \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t \right]^{-1} (1/T) \left[\sum_{t=1}^T \hat{u}_t^2 \mathbf{x}_t \mathbf{x}'_t \right. \\ &\quad \left. + \sum_{v=1}^q \left[1 - \frac{v}{q+1} \right] \sum_{t=v+1}^T (\mathbf{x}_t \hat{u}_t \hat{u}_{t-v} \mathbf{x}'_{t-v} + \mathbf{x}_{t-v} \hat{u}_{t-v} \hat{u}_t \mathbf{x}'_t) \right] \\ &\quad \times \left[(1/T) \sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t \right]^{-1}, \end{aligned}$$

that is, the variance of $\mathbf{b}_T$ is approximated by

$$\begin{aligned} &(\hat{\mathbf{V}}_T/T) \\ &= \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t \right]^{-1} \left[\sum_{t=1}^T \hat{u}_t^2 \mathbf{x}_t \mathbf{x}'_t \right. \\ &\quad \left. + \sum_{v=1}^q \left[1 - \frac{v}{q+1} \right] \sum_{t=v+1}^T (\mathbf{x}_t \hat{u}_t \hat{u}_{t-v} \mathbf{x}'_{t-v} + \mathbf{x}_{t-v} \hat{u}_{t-v} \hat{u}_t \mathbf{x}'_t) \right] \left[\sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t \right]^{-1} \end{aligned} \quad [10.5.21]$$

where $\hat{u}_t$ is the *OLS* sample residual. The square root of the row i , column i element of $\hat{\mathbf{V}}_T/T$ is known as a *heteroskedasticity- and autocorrelation-consistent standard error* for the i th element of the estimated *OLS* coefficient vector. The hope is that standard errors based on [10.5.21] will be robust to a variety of forms of heteroskedasticity and autocorrelation of the residuals u_t of the regression.

Spectral-Based Estimators

A number of alternative estimates of $\mathbf{S}$ in [10.5.7] have been suggested in the literature. Notice that as in the univariate case discussed in Section 7.2, if $\mathbf{y}_t$ is covariance-stationary, then $\mathbf{S}$ has the interpretation as the autocovariance-generating function $\mathbf{G}_{\mathbf{Y}}(z) = \sum_{v=-\infty}^{\infty} \boldsymbol{\Gamma}_v z^v$ evaluated at $z = 1$, or, equivalently, as 2π times the population spectrum at frequency zero:

$$\mathbf{S} = \sum_{v=-\infty}^{\infty} \boldsymbol{\Gamma}_v = 2\pi \mathbf{s}_{\mathbf{Y}}(0).$$

Indeed, the Newey-West estimator [10.5.15] is numerically identical to 2π times the Bartlett estimate of the multivariate spectrum described in [10.4.42] evaluated at frequency $\omega = 0$. Gallant (1987, p. 533) proposed a similar estimator based on a Parzen kernel,

$$\hat{\mathbf{S}} = \hat{\mathbf{r}}_0 + \sum_{v=1}^q k[v/(q+1)](\hat{\mathbf{r}}_v + \hat{\mathbf{r}}'_v),$$

where

$$k(z) = \begin{cases} 1 - 6z^2 + 6z^3 & \text{for } 0 \leq z \leq \frac{1}{2} \\ 2(1 - z)^3 & \text{for } \frac{1}{2} \leq z \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

For example, for $q = 2$, we have

$$\hat{\mathbf{S}} = \hat{\mathbf{r}}_0 + \frac{5}{6}(\hat{\mathbf{r}}_1 + \hat{\mathbf{r}}_1') + \frac{2}{3}(\hat{\mathbf{r}}_2 + \hat{\mathbf{r}}_2').$$

Andrews (1991) examined a number of alternative estimators and found the best results for a quadratic spectral kernel:

$$k(z) = \frac{3}{(6\pi z/5)^2} \left[\frac{\sin(6\pi z/5)}{6\pi z/5} - \cos(6\pi z/5) \right].$$

In contrast to the Newey-West and Gallant estimators, Andrews's suggestion makes use of all $T - 1$ estimated autocovariance estimators:

$$\hat{\mathbf{S}} = \frac{T}{T - k} \left[\hat{\mathbf{r}}_0 + \sum_{v=1}^{T-1} k\left(\frac{v}{q+1}\right)(\hat{\mathbf{r}}_v + \hat{\mathbf{r}}_v') \right]. \quad [10.5.22]$$

Even though [10.5.22] makes use of all computed autocovariances, there is still a bandwidth parameter q to be chosen for constructing the kernel. For example, for $q = 2$,

$$\begin{aligned} \hat{\mathbf{r}}_0 + \sum_{v=1}^{T-1} k(v/3)(\hat{\mathbf{r}}_v + \hat{\mathbf{r}}_v') &= \hat{\mathbf{r}}_0 + 0.85(\hat{\mathbf{r}}_1 + \hat{\mathbf{r}}_1') \\ &\quad + 0.50(\hat{\mathbf{r}}_2 + \hat{\mathbf{r}}_2') + 0.14(\hat{\mathbf{r}}_3 + \hat{\mathbf{r}}_3') + \cdots \end{aligned}$$

Andrews recommended multiplying the estimate by $T/(T - k)$, where $\mathbf{y}_t = \mathbf{x}_t \hat{\mathbf{u}}_t$, for $\hat{\mathbf{u}}_t$ the sample *OLS* residual from a regression with k explanatory variables. Andrews (1991) and Newey and West (1994) also offered some guidance for choosing an optimal value of the lag truncation or bandwidth parameter q for each of the estimators of $\mathbf{S}$ that have been discussed here.

The estimators that have been described will work best when $\mathbf{y}_t$ has a finite moving average representation. Andrews and Monahan (1992) suggested an alternative approach to estimating $\mathbf{S}$ that also takes advantage of any autoregressive structure to the errors. Let $\mathbf{y}_t$ be a zero-mean vector, and let $\mathbf{S}$ be the asymptotic variance of the sample mean of $\mathbf{y}$. For example, if we want to calculate heteroskedasticity- and autocorrelation-consistent standard errors for *OLS* estimation, $\mathbf{y}_t$ would correspond to $\mathbf{x}_t \hat{\mathbf{u}}_t$, where $\mathbf{x}_t$ is the vector of explanatory variables for the regression and $\hat{\mathbf{u}}_t$ is the *OLS* residual. The first step in estimating $\mathbf{S}$ is to fit a low-order *VAR* for $\mathbf{y}_t$,

$$\mathbf{y}_t = \Phi_1 \mathbf{y}_{t-1} + \Phi_2 \mathbf{y}_{t-2} + \cdots + \Phi_p \mathbf{y}_{t-p} + \mathbf{v}_t, \quad [10.5.23]$$

where $\mathbf{v}_t$ is presumed to have some residual autocorrelation not entirely captured by the *VAR*. Note that since $\mathbf{y}_t$ has mean zero, no constant term is included in [10.5.23]. The i th row represented in [10.5.23] can be estimated by an *OLS* regression of the i th element of $\mathbf{y}_t$ on p lags of all the elements of $\mathbf{y}$, though if any eigenvalue of $|\mathbf{I}_n \lambda^p - \Phi_1 \lambda^{p-1} - \Phi_2 \lambda^{p-2} - \cdots - \Phi_p| = 0$ is too close to the unit circle (say, greater than 0.97 in modulus), Andrews and Monahan (1992, p. 957) recommended altering the *OLS* estimates so as to reduce the largest eigenvalue.

The second step in the Andrews and Monahan procedure is to calculate an estimate $\mathbf{S}^*$ using one of the methods described previously based on the fitted

residuals $\hat{\mathbf{v}}_t$ from [10.5.23]. For example,

$$\hat{\mathbf{S}}_T^* = \hat{\mathbf{\Gamma}}_0^* + \sum_{v=1}^q \left[1 - \frac{v}{q+1} \right] (\hat{\mathbf{\Gamma}}_v^* + \hat{\mathbf{\Gamma}}_v^{*'}), \quad [10.5.24]$$

where

$$\hat{\mathbf{\Gamma}}_v^* = (1/T) \sum_{t=v+1}^T \hat{\mathbf{v}}_t \hat{\mathbf{v}}_{t-v}'$$

and where q is a parameter representing the maximal order of autocorrelation assumed for $\mathbf{v}_t$. The matrix $\hat{\mathbf{S}}_T^*$ will be recognized as an estimate of $2\pi \cdot \mathbf{s}_v(0)$, where $\mathbf{s}_v(\omega)$ is the spectral density of $\mathbf{v}$:

$$\mathbf{s}_v(\omega) = (2\pi)^{-1} \sum_{v=-\infty}^{\infty} \{E(\mathbf{v}_t \mathbf{v}_{t-v}')\} e^{-i\omega v}.$$

Notice that the original series $\mathbf{y}_t$ can be obtained from $\mathbf{v}_t$ by applying the following filter:

$$\mathbf{y}_t = [\mathbf{I}_n - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p]^{-1} \mathbf{v}_t.$$

Thus, from [10.4.43], the spectral density of $\mathbf{y}$ is related to the spectral density of $\mathbf{v}$ according to

$$\begin{aligned} \mathbf{s}_y(\omega) &= \{[\mathbf{I}_n - \Phi_1 e^{-i\omega} - \Phi_2 e^{-2i\omega} - \cdots - \Phi_p e^{-pi\omega}]\}^{-1} \mathbf{s}_v(\omega) \\ &\quad \times \{[\mathbf{I}_n - \Phi_1 e^{i\omega} - \Phi_2 e^{2i\omega} - \cdots - \Phi_p e^{pi\omega}]\}'^{-1}. \end{aligned}$$

Hence, an estimate of 2π times the spectral density of $\mathbf{y}$ at frequency zero is given by

$$\begin{aligned} \hat{\mathbf{S}}_T &= \{[\mathbf{I}_n - \hat{\Phi}_1 - \hat{\Phi}_2 - \cdots - \hat{\Phi}_p]\}^{-1} \hat{\mathbf{S}}_T^* \\ &\quad \times \{[\mathbf{I}_n - \hat{\Phi}_1 - \hat{\Phi}_2 - \cdots - \hat{\Phi}_p]\}'^{-1}, \end{aligned} \quad [10.5.25]$$

where $\hat{\mathbf{S}}_T^*$ is calculated from [10.5.24]. The matrix $\hat{\mathbf{S}}_T$ in [10.5.25] is the Andrews-Monahan (1992) estimate of $\mathbf{S}$, where

$$\mathbf{S} = \lim_{T \rightarrow \infty} T \cdot E(\bar{\mathbf{y}}_T \bar{\mathbf{y}}_T').$$

APPENDIX 10.A. Proofs of Chapter 10 Propositions

■ **Proof of Proposition 10.1.** The eigenvalues of $\mathbf{F}$ are the values of λ for which the following determinant is zero:

$$\begin{vmatrix} (\Phi_1 - \lambda \mathbf{I}_n) & \Phi_2 & \Phi_3 & \cdots & \Phi_{p-1} & \Phi_p \\ \mathbf{I}_n & -\lambda \mathbf{I}_n & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_n & -\lambda \mathbf{I}_n & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_n & -\lambda \mathbf{I}_n \end{vmatrix}. \quad [10.A.1]$$

Multiply each of the final block of n columns by $(1/\lambda)$ and add to the previous block. Multiply each of the n columns of this resulting next-to-final block by $(1/\lambda)$ and add the result to the third-to-last block of columns. Proceeding in this manner reveals [10.A.1] to be the same as

$$\begin{vmatrix} \mathbf{X}_1 & \mathbf{X}_2 \\ \mathbf{0} & -\lambda \mathbf{I}_{n(p-1)} \end{vmatrix}, \quad [10.A.2]$$

where $\mathbf{X}_1$ denotes the following $(n \times n)$ matrix:

$$\mathbf{X}_1 \equiv (\Phi_1 - \lambda \mathbf{I}_n) + (\Phi_2/\lambda) + (\Phi_3/\lambda^2) + \cdots + (\Phi_p/\lambda^{p-1})$$

and $\mathbf{X}_2$ is a related $[n \times n(p-1)]$ matrix. Let $\mathbf{S}$ denote the following $(np \times np)$ matrix:

$$\mathbf{S} \equiv \begin{bmatrix} \mathbf{0} & \mathbf{I}_{n(p-1)} \\ \mathbf{I}_n & \mathbf{0} \end{bmatrix},$$

and note that its inverse is given by

$$\mathbf{S}^{-1} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_n \\ \mathbf{I}_{n(p-1)} & \mathbf{0} \end{bmatrix},$$

as may be verified by direct multiplication. Premultiplying a matrix by $\mathbf{S}$ and postmultiplying by $\mathbf{S}^{-1}$ will not change the determinant. Thus, [10.A.2] is equal to

$$\left| \begin{bmatrix} \mathbf{0} & \mathbf{I}_{n(p-1)} \\ \mathbf{I}_n & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 \\ \mathbf{0} & -\lambda \mathbf{I}_{n(p-1)} \end{bmatrix} \begin{bmatrix} \mathbf{0} & \mathbf{I}_n \\ \mathbf{I}_{n(p-1)} & \mathbf{0} \end{bmatrix} \right| = \left| \begin{bmatrix} -\lambda \mathbf{I}_{n(p-1)} & \mathbf{0} \\ \mathbf{X}_2 & \mathbf{X}_1 \end{bmatrix} \right|. \quad [10.A.3]$$

Applying the formula for calculating a determinant [A.4.5] recursively, [10.A.3] is equal to

$$\begin{aligned} (-\lambda)^{n(p-1)} |\mathbf{X}_1| &= (-\lambda)^{n(p-1)} [\Phi_1 - \lambda \mathbf{I}_n + (\Phi_2/\lambda) + (\Phi_3/\lambda^2) + \cdots + (\Phi_p/\lambda^{p-1})] \\ &= (-1)^{np} [\mathbf{I}_n \lambda^p - \Phi_1 \lambda^{p-1} - \Phi_2 \lambda^{p-2} - \cdots - \Phi_p]. \end{aligned}$$

Setting this to zero produces equation [10.1.13]. ■

■ **Proof of Proposition 10.2.** It is helpful to define $z_i(i, l)$ to be the component of y_{it} that reflects the cumulative effects of the l th element of $\boldsymbol{\varepsilon}$:

$$z_i(i, l) = \psi_{il}^{(0)} \varepsilon_{it} + \psi_{il}^{(1)} \varepsilon_{i,t-1} + \psi_{il}^{(2)} \varepsilon_{i,t-2} + \cdots = \sum_{v=0}^{\infty} \psi_{il}^{(v)} \varepsilon_{i,t-v}, \quad [10.A.4]$$

where $\psi_{il}^{(v)}$ denotes the row i , column l element of the matrix $\boldsymbol{\Psi}_v$. The actual value of the i th variable y_{it} is the sum of the contributions of each of the $l = 1, 2, \dots, n$ components of $\boldsymbol{\varepsilon}$:

$$y_{it} = \mu_i + \sum_{l=1}^n z_i(i, l). \quad [10.A.5]$$

The results of Proposition 10.2 are all established by first demonstrating absolute summability of the moments of $z_i(i, l)$ and then observing that the moments of y , are obtained from finite sums of these expressions based on $z_i(i, l)$.

Proof of (a). Consider the random variable $z_i(i, l) \cdot z_{i-s}(j, m)$, where i, l, j , and m represent arbitrary indices between 1 and n and where s is the order of the autocovariance of y that is being calculated. Note from [10.A.4] that

$$\begin{aligned} E\{z_i(i, l) \cdot z_{i-s}(j, m)\} &= E\left\{\left[\sum_{r=0}^{\infty} \psi_{il}^{(r)} \varepsilon_{i,t-r}\right] \times \left[\sum_{v=0}^{\infty} \psi_{jm}^{(v)} \varepsilon_{m,t-s-v}\right]\right\} \\ &= \sum_{r=0}^{\infty} \sum_{v=0}^{\infty} \{\psi_{il}^{(r)} \psi_{jm}^{(v)}\} \cdot E\{\varepsilon_{i,t-r} \varepsilon_{m,t-s-v}\}. \end{aligned} \quad [10.A.6]$$

The expectation operator can be moved inside the summation here because

$$\sum_{r=0}^{\infty} \sum_{v=0}^{\infty} |\psi_{il}^{(r)} \psi_{jm}^{(v)}| = \sum_{r=0}^{\infty} \sum_{v=0}^{\infty} |\psi_{il}^{(r)}| \cdot |\psi_{jm}^{(v)}| = \left\{ \sum_{r=0}^{\infty} |\psi_{il}^{(r)}| \right\} \times \left\{ \sum_{v=0}^{\infty} |\psi_{jm}^{(v)}| \right\} < \infty.$$

Now, the product of ε 's in the final term in [10.A.6] can have nonzero expectation only if the ε 's have the same date, that is, if $r = s + v$. Thus, although [10.A.6] involves a sum over an infinite number of values of r , only the value at $r = s + v$ contributes to this sum:

$$E\{z_i(i, l) \cdot z_{i-s}(j, m)\} = \sum_{v=0}^{\infty} \{\psi_{il}^{(s+v)} \psi_{jm}^{(v)}\} \cdot E\{\varepsilon_{i,t-s-v} \varepsilon_{m,t-s-v}\} = \sum_{v=0}^{\infty} \psi_{il}^{(s+v)} \psi_{jm}^{(v)} \sigma_{lm}, \quad [10.A.7]$$

where σ_{lm} represents the covariance between ε_{it} and ε_{mt} and is given by the row l , column m element of $\boldsymbol{\Omega}$.

The row i , column j element of $\boldsymbol{\Gamma}_s$ gives the value of

$$\gamma_{ij}^{(s)} = E(y_{it} - \mu_i)(y_{j,t-s} - \mu_j).$$

Using [10.A.5] and [10.A.7], this can be expressed as

$$\begin{aligned}
 E(y_{it} - \mu_i)(y_{j,t-s} - \mu_j) &= E \left\{ \left[\sum_{l=1}^n z_t(i, l) \right] \left[\sum_{m=1}^n z_{t-s}(j, m) \right] \right\} \\
 &= \sum_{l=1}^n \sum_{m=1}^n E \{ z_t(i, l) \cdot z_{t-s}(j, m) \} \\
 &= \sum_{l=1}^n \sum_{m=1}^n \sum_{v=0}^{\infty} \psi_{il}^{(s+v)} \psi_{jm}^{(v)} \sigma_{lm} \\
 &= \sum_{v=0}^{\infty} \sum_{l=1}^n \sum_{m=1}^n \psi_{il}^{(s+v)} \psi_{jm}^{(v)} \sigma_{lm}.
 \end{aligned} \tag{10.A.8}$$

But $\sum_{l=1}^n \sum_{m=1}^n \psi_{il}^{(s+v)} \psi_{jm}^{(v)} \sigma_{lm}$ is the row i , column j element of $\Psi_{v+s} \Omega \Psi'_v$. Thus, [10.A.8] states that the row i , column j element of Γ_s is given by the row i , column j element of $\sum_{v=0}^{\infty} \Psi_{v+s} \Omega \Psi'_v$, as asserted in part (a).

Proof of (b). Define $h_s(\cdot)$ to be the moment in [10.A.7]:

$$h_s(i, j, l, m) \equiv E \{ z_t(i, l) \cdot z_{t-s}(j, m) \} = \sum_{v=0}^{\infty} \psi_{il}^{(s+v)} \psi_{jm}^{(v)} \sigma_{lm};$$

and notice that the sequence $\{h_s(\cdot)\}_{s=0}^{\infty}$ is absolutely summable:

$$\begin{aligned}
 \sum_{s=0}^{\infty} |h_s(i, j, l, m)| &\leq \sum_{s=0}^{\infty} \sum_{v=0}^{\infty} |\psi_{il}^{(s+v)}| \cdot |\psi_{jm}^{(v)}| \cdot |\sigma_{lm}| \\
 &= |\sigma_{lm}| \sum_{v=0}^{\infty} |\psi_{jm}^{(v)}| \sum_{s=0}^{\infty} |\psi_{il}^{(s+v)}| \\
 &\leq |\sigma_{lm}| \sum_{v=0}^{\infty} |\psi_{jm}^{(v)}| \sum_{s=0}^{\infty} |\psi_{il}^{(s)}| \\
 &< \infty.
 \end{aligned} \tag{10.A.9}$$

Furthermore, the row i , column j element of Γ_s was seen in [10.A.8] to be given by

$$\gamma_{ij}^{(s)} = \sum_{l=1}^n \sum_{m=1}^n h_s(i, j, l, m).$$

Hence,

$$\sum_{s=0}^{\infty} |\gamma_{ij}^{(s)}| \leq \sum_{s=0}^{\infty} \sum_{l=1}^n \sum_{m=1}^n |h_s(i, j, l, m)| = \sum_{l=1}^n \sum_{m=1}^n \sum_{s=0}^{\infty} |h_s(i, j, l, m)|. \tag{10.A.10}$$

From [10.A.9], there exists an $M < \infty$ such that

$$\sum_{s=0}^{\infty} |h_s(i, j, l, m)| < M$$

for any value of i, j, l , or m . Hence, [10.A.10] implies

$$\sum_{s=0}^{\infty} |\gamma_{ij}^{(s)}| < \sum_{l=1}^n \sum_{m=1}^n M = n^2 M < \infty,$$

confirming that the row i , column j element of $\{\Gamma_s\}_{s=0}^{\infty}$ is absolutely summable, as claimed by part (b).

Proof of (c). Essentially the identical algebra as in the proof of Proposition 7.10 establishes that

$$\begin{aligned}
 & E|z_{i_1}(i_1, l_1) \cdot z_{i_2}(i_2, l_2) \cdot z_{i_3}(i_3, l_3) \cdot z_{i_4}(i_4, l_4)| \\
 &= E \left| \left\{ \sum_{v_1=0}^{\infty} \psi_{i_1, l_1}^{(v_1)} \varepsilon_{i_1, l_1 - v_1} \right\} \cdot \left\{ \sum_{v_2=0}^{\infty} \psi_{i_2, l_2}^{(v_2)} \varepsilon_{i_2, l_2 - v_2} \right\} \right. \\
 &\quad \cdot \left\{ \sum_{v_3=0}^{\infty} \psi_{i_3, l_3}^{(v_3)} \varepsilon_{i_3, l_3 - v_3} \right\} \cdot \left\{ \sum_{v_4=0}^{\infty} \psi_{i_4, l_4}^{(v_4)} \varepsilon_{i_4, l_4 - v_4} \right\} \Big| \quad [10.A.11] \\
 &\leq \sum_{v_1=0}^{\infty} \sum_{v_2=0}^{\infty} \sum_{v_3=0}^{\infty} \sum_{v_4=0}^{\infty} |\psi_{i_1, l_1}^{(v_1)} \psi_{i_2, l_2}^{(v_2)} \psi_{i_3, l_3}^{(v_3)} \psi_{i_4, l_4}^{(v_4)}| \\
 &\quad \times E|\varepsilon_{i_1, l_1 - v_1} \varepsilon_{i_2, l_2 - v_2} \varepsilon_{i_3, l_3 - v_3} \varepsilon_{i_4, l_4 - v_4}| \\
 &< \infty.
 \end{aligned}$$

Now,

$$\begin{aligned}
 E|y_{i_1, l_1} y_{i_2, l_2} y_{i_3, l_3} y_{i_4, l_4}| &= E \left| \mu_{i_1} + \sum_{l_1=1}^n z_{i_1}(i_1, l_1) \right| \cdot \left| \mu_{i_2} + \sum_{l_2=1}^n z_{i_2}(i_2, l_2) \right| \\
 &\quad \cdot \left| \mu_{i_3} + \sum_{l_3=1}^n z_{i_3}(i_3, l_3) \right| \cdot \left| \mu_{i_4} + \sum_{l_4=1}^n z_{i_4}(i_4, l_4) \right| \\
 &\leq E \left\{ |\mu_{i_1}| + \sum_{l_1=1}^n |z_{i_1}(i_1, l_1)| \right\} \cdot \left\{ |\mu_{i_2}| + \sum_{l_2=1}^n |z_{i_2}(i_2, l_2)| \right\} \\
 &\quad \cdot \left\{ |\mu_{i_3}| + \sum_{l_3=1}^n |z_{i_3}(i_3, l_3)| \right\} \cdot \left\{ |\mu_{i_4}| + \sum_{l_4=1}^n |z_{i_4}(i_4, l_4)| \right\}.
 \end{aligned}$$

But this is a finite sum involving terms of the form of [10.A.11]—which were seen to be finite—along with terms involving first through third moments of z , which must also be finite.

Proof of (d). Notice that

$$z_i(i, l) \cdot z_{i-s}(j, m) = \sum_{r=0}^{\infty} \sum_{v=0}^{\infty} \psi_{i,l}^{(r)} \psi_{i-m}^{(v)} \varepsilon_{i, l-r} \varepsilon_{i-m, m-v}.$$

The same argument leading to [7.2.14] can be used to establish that

$$(1/T) \sum_{i=1}^T z_i(i, l) \cdot z_{i-s}(j, m) \xrightarrow{p} E\{z_i(i, l) \cdot z_{i-s}(j, m)\}. \quad [10.A.12]$$

To see that [10.A.12] implies ergodicity for the second moments of y , notice from [10.A.5] that

$$\begin{aligned}
 (1/T) \sum_{i=1}^T y_{it} y_{i-t, s} &= (1/T) \sum_{i=1}^T \left[\mu_i + \sum_{l=1}^n z_i(i, l) \right] \left[\mu_{i-s} + \sum_{m=1}^n z_{i-s}(j, m) \right] \\
 &= \mu_i \mu_{i-s} + \mu_i \sum_{m=1}^n \left[(1/T) \sum_{i=1}^T z_{i-s}(j, m) \right] + \mu_{i-s} \sum_{l=1}^n \left[(1/T) \sum_{i=1}^T z_i(i, l) \right] \\
 &\quad + \sum_{l=1}^n \sum_{m=1}^n \left[(1/T) \sum_{i=1}^T z_i(i, l) z_{i-s}(j, m) \right] \\
 &\xrightarrow{p} \mu_i \mu_{i-s} + \mu_i \sum_{m=1}^n E[z_{i-s}(j, m)] + \mu_{i-s} \sum_{l=1}^n E[z_i(i, l)] \\
 &\quad + \sum_{l=1}^n \sum_{m=1}^n E[z_i(i, l) z_{i-s}(j, m)] \\
 &= E \left\{ \left[\mu_i + \sum_{l=1}^n z_i(i, l) \right] \left[\mu_{i-s} + \sum_{m=1}^n z_{i-s}(j, m) \right] \right\} \\
 &= E[y_{it} y_{i-t, s}],
 \end{aligned}$$

as claimed. ■

■ **Proof of Proposition 10.3.** Writing out [10.2.11] explicitly,

$$\mathbf{H}(L)\Psi(L) = (\cdots + \mathbf{H}_{-1}L^{-1} + \mathbf{H}_0L^0 + \mathbf{H}_1L^1 + \cdots) \\ \times (\Psi_0L^0 + \Psi_1L^1 + \Psi_2L^2 + \cdots),$$

from which the coefficient on L^k is

$$\mathbf{B}_k = \mathbf{H}_k\Psi_0 + \mathbf{H}_{k-1}\Psi_1 + \mathbf{H}_{k-2}\Psi_2 + \cdots. \quad [10.A.13]$$

Let $b_{ij}^{(k)}$ denote the row i , column j element of $\mathbf{B}_k$, and let $h_{im}^{(k)}$ and $\psi_{mj}^{(k)}$ denote the row i , column j elements of $\mathbf{H}_k$ and Ψ_k , respectively. Then the row i , column j element of the matrix equation [10.A.13] states that

$$b_{ij}^{(k)} = \sum_{m=1}^n h_{im}^{(k)}\psi_{mj}^{(0)} + \sum_{m=1}^n h_{im}^{(k-1)}\psi_{mj}^{(1)} + \sum_{m=1}^n h_{im}^{(k-2)}\psi_{mj}^{(2)} + \cdots = \sum_{v=0}^{\infty} \sum_{m=1}^n h_{im}^{(k-v)}\psi_{mj}^{(v)}.$$

Thus,

$$\begin{aligned} \sum_{k=-\infty}^{\infty} |b_{ij}^{(k)}| &= \sum_{k=-\infty}^{\infty} \left| \sum_{v=0}^{\infty} \sum_{m=1}^n h_{im}^{(k-v)}\psi_{mj}^{(v)} \right| \\ &\leq \sum_{k=-\infty}^{\infty} \sum_{v=0}^{\infty} \sum_{m=1}^n |h_{im}^{(k-v)}\psi_{mj}^{(v)}| \\ &= \sum_{m=1}^n \sum_{v=0}^{\infty} |\psi_{mj}^{(v)}| \sum_{k=-\infty}^{\infty} |h_{im}^{(k-v)}|. \end{aligned} \quad [10.A.14]$$

But since $\{\mathbf{H}_k\}_k$ and $\{\Psi_k\}_k$ are absolutely summable,

$$\begin{aligned} \sum_{k=-\infty}^{\infty} |h_{im}^{(k-v)}| &< M_1 < \infty \\ \sum_{v=0}^{\infty} |\psi_{mj}^{(v)}| &< M_2 < \infty. \end{aligned}$$

Thus, [10.A.14] becomes

$$\sum_{k=-\infty}^{\infty} |b_{ij}^{(k)}| < \sum_{m=1}^n M_1 M_2 < \infty. \quad \blacksquare$$

■ **Proof of Proposition 10.4.** Let $\mathbf{A}$ be $(m \times n)$, $\mathbf{B}$ be $(n \times r)$, and $\mathbf{C}$ be $(r \times q)$. Let the $(n \times 1)$ vector $\mathbf{b}_i$ denote the i th column of $\mathbf{B}$, and let c_{ij} denote the row i , column j element of $\mathbf{C}$. Then

$$\begin{aligned} \mathbf{ABC} &= \mathbf{A}[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_r] \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1q} \\ c_{21} & c_{22} & \cdots & c_{2q} \\ \vdots & \vdots & \cdots & \vdots \\ c_{r1} & c_{r2} & \cdots & c_{rq} \end{bmatrix} \\ &= \{ \{\mathbf{Ab}_1 c_{11} + \mathbf{Ab}_2 c_{21} + \cdots + \mathbf{Ab}_r c_{r1}\} \\ &\quad \{\mathbf{Ab}_1 c_{12} + \mathbf{Ab}_2 c_{22} + \cdots + \mathbf{Ab}_r c_{r2}\} \cdots \\ &\quad \{\mathbf{Ab}_1 c_{1q} + \mathbf{Ab}_2 c_{2q} + \cdots + \mathbf{Ab}_r c_{rq}\} \} \\ &= \{ \{c_{11}\mathbf{Ab}_1 + c_{21}\mathbf{Ab}_2 + \cdots + c_{r1}\mathbf{Ab}_r\} \\ &\quad \{c_{12}\mathbf{Ab}_1 + c_{22}\mathbf{Ab}_2 + \cdots + c_{r2}\mathbf{Ab}_r\} \cdots \\ &\quad \{c_{1q}\mathbf{Ab}_1 + c_{2q}\mathbf{Ab}_2 + \cdots + c_{rq}\mathbf{Ab}_r\} \}. \end{aligned}$$

Applying the vec operator gives

$$\begin{aligned} \text{vec}(\mathbf{ABC}) &= \begin{bmatrix} c_{11}\mathbf{Ab}_1 + c_{21}\mathbf{Ab}_2 + \cdots + c_{r1}\mathbf{Ab}_r \\ c_{12}\mathbf{Ab}_1 + c_{22}\mathbf{Ab}_2 + \cdots + c_{r2}\mathbf{Ab}_r \\ \vdots \\ c_{1q}\mathbf{Ab}_1 + c_{2q}\mathbf{Ab}_2 + \cdots + c_{rq}\mathbf{Ab}_r \end{bmatrix} \\ &= \begin{bmatrix} c_{11}\mathbf{A} & c_{21}\mathbf{A} & \cdots & c_{r1}\mathbf{A} \\ c_{12}\mathbf{A} & c_{22}\mathbf{A} & \cdots & c_{r2}\mathbf{A} \\ \vdots & \vdots & \cdots & \vdots \\ c_{1q}\mathbf{A} & c_{2q}\mathbf{A} & \cdots & c_{rq}\mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_r \end{bmatrix} \\ &= (\mathbf{C}' \otimes \mathbf{A}) \cdot \text{vec}(\mathbf{B}). \quad \blacksquare \end{aligned}$$

Chapter 10 Exercises

10.1. Consider a scalar $AR(p)$ process ($n = 1$). Deduce from equation [10.2.19] that the $(p \times 1)$ vector consisting of the variance and first $(p - 1)$ autocovariances,

$$\begin{bmatrix} \gamma_0 \\ \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_{p-1} \end{bmatrix},$$

can be calculated from the first p elements in the first column of the $(p^2 \times p^2)$ matrix $\sigma^2[\mathbf{I}_{p^2} - (\mathbf{F} \otimes \mathbf{F})]^{-1}$ for $\mathbf{F}$ the $(p \times p)$ matrix defined in equation [1.2.3] in Chapter 1.

10.2. Let $\mathbf{y}_t = (X_t, Y_t)'$ be given by

$$\begin{aligned} X_t &= \varepsilon_t + \theta \varepsilon_{t-1} \\ Y_t &= h_1 X_{t-1} + u_t, \end{aligned}$$

where $(\varepsilon_t, u_t)'$ is vector white noise with contemporaneous variance-covariance matrix given by

$$\begin{bmatrix} E(\varepsilon_t^2) & E(\varepsilon_t u_t) \\ E(u_t \varepsilon_t) & E(u_t^2) \end{bmatrix} = \begin{bmatrix} \sigma_\varepsilon^2 & 0 \\ 0 & \sigma_u^2 \end{bmatrix}.$$

(a) Calculate the autocovariance matrices $\{\Gamma_k\}_{k=-\infty}^{\infty}$ for this process.

(b) Use equation [10.4.3] to calculate the population spectrum. Find the cospectrum between X and Y and the quadrature spectrum from X to Y .

(c) Verify that your answer to part (b) could equivalently be calculated from expression [10.4.45].

(d) Verify by integrating your answer to part (b) that [10.4.49] holds; that is, show that

$$(2\pi)^{-1} \int_{-\pi}^{\pi} \frac{s_{YX}(\omega)}{\pi s_{XX}(\omega)} e^{i\omega k} d\omega = \begin{cases} h_1 & \text{for } k = 1 \\ 0 & \text{for other integer } k. \end{cases}$$

Chapter 10 References

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